

The evolutionary interplay between self-fertilisation and sex allocation in flowering plants

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Abstract. Flowering plants vary in sex allocation (resource division between male and female function) and in selfing rate. These traits should evolve together: selfing changes fitness returns from pollen and ovules, whereas sex allocation changes the pollen available for selfing and export. Yet most models allow only one trait to evolve, leaving their evolutionary feedback unresolved. Here we model the joint evolution of sex allocation, pollen export and maternal control of competition between self- and cross-pollen. We show that selfing and sex allocation strongly influence one another, as selection favours associations between sex allocation and mating-system traits. When maternal control is fixed to recover classical mass-action, evolution can lead to the stable coexistence of selfing hermaphrodites and males specialised in pollen export. This androdioecy persists because hermaphrodites cannot prevent fertilisation by males. When maternal control of pollen competition also evolves, we find that where a mixed-mating equilibrium exists, selection again favours the emergence of androdioecy. As hermaphrodites evolve to favour their own pollen, however, they progressively exclude pollen from males, driving them extinct. The population may then return to mixed mating and repeat the process, or evolve complete selfing. We discuss these results with reference to known variation in the mating and sexual systems of plants.

Keywords. Mating systems, Sexual systems, Hermaphrodites, Androdioecy, Pollen competition, Evolutionary branching

1 Introduction

Flowering plants vary widely in their sexual system, that is in how male and female functions are distributed across individuals (Renner and Ricklefs, 1995; Renner, 2014; Käfer et al., 2017). Most species are hermaphroditic, meaning that each individual produces both pollen and ovules, but the resources allocated to each function can still vary substantially (Cruden, 1977; Cruden and Lyon, 1985; Brunet, 1992; Jürgens et al., 2002). Other species have evolved polymorphic sexual systems including dioecy, in which individuals are either male or female, as well as rarer systems in which hermaphroditic and unisexual individuals (males or females) coexist in the same population (Renner, 2014; Käfer et al., 2017). This diversity reflects the range of selective pressures acting on plant sex allocation, i.e., how plants divide their reproductive resources between male and female function (Freeman et al., 1997; Käfer et al., 2017; Pannell and Jordan, 2022).

The evolution of plant sex allocation depends on ecological features that influence the fitness returns from producing pollen and ovules (Charnov et al., 1976; Charnov, 1982). These features for example include the density of potential mates and competitors through each function, plant size, and interactions with animal pollinators, seed dispersers and predators (for reviews, see Bawa, 1980; Charlesworth, 1993, 1999; Freeman et al., 1997; De Jong and Klinkhamer, 2005; Barrett and Harder, 2017; Leonard, 2018; Pannell and Jordan, 2022). Among them, the population's mating system – the dynamics of pollen exchange between flowers that determine who fertilises whom (Barrett and Harder, 2017) – is especially influential.

Self-fertilisation (selfing hereafter), a common mating strategy in hermaphrodites whereby individuals fertilise their own ovules, is thought to have profound consequences for sex allocation evolution. First, selfing changes expected fitness returns through male and female function in a way that favours female-biased sex allocation (Charlesworth and Charlesworth, 1981; Zhang, 2000; Lesaffre et al., 2025). This is because selfing simultaneously : (i) decreases expected fitness gains as a pollen donor by reducing the number of ovules available for outcrossing; and (ii) increases expected fitness returns through seeds when inbreeding depression – the decrease in relative fitness of inbred progeny owing to homozygosity for recessive deleterious mutations (Charlesworth and Charlesworth, 1987) – is weak relative to the trans-

mission advantage that selfing entails (a selfing parent acts as both mother and father to its offspring, and so transmits twice as many gene copies per seed on average; Fisher, 1941). Second, selfing is thought to play a central role in many evolutionary transitions from combined to separate sexes that have occurred in flowering plants (Charlesworth, 1999; De Jong and Geritz, 2001; Pannell and Jordan, 2022). Models show that selfing and inbreeding depression can facilitate the evolution of separate sexes, either through the step-wise invasion of hermaphroditic populations by sterility mutations creating unisexual females and males (Charlesworth and Charlesworth, 1978), or via the gradual emergence of specialised male and female morphs (Lesaffre et al., 2024).

Existing theory has clarified how selfing affects sex allocation evolution, but most of it treats the mating system as fixed, despite the fact that selfing rates are known to evolve rapidly (Takebayashi and Morell, 2001; Goldberg et al., 2010; Barrett and Harder, 2017). The mechanisms shaping selfing evolution have been extensively discussed (Goodwillie et al., 2005; Clo et al., 2025). Early theoretical treatments show that the opposition between the transmission advantage (Fisher, 1941) and inbreeding depression generates a threshold effect: selection favours either complete outcrossing or complete selfing depending on the severity of inbreeding depression, so that mixed mating, where individuals self-fertilise at an intermediate rate, should not be stable (Lande and Schemske, 1985). This insight came from models treating inbreeding depression as a parameter, but studies that explicitly modelled the genetic basis of inbreeding depression reached similar conclusions (e.g., Charlesworth et al., 1990, 1991; Porcher et al., 2009; Lande and Porcher, 2015; Abu Awad and Billiard, 2017; Abu Awad and Roze, 2020). Additional mechanisms may therefore be needed to explain the widespread occurrence of mixed mating in flowering plants (Goodwillie et al., 2005; Whitehead et al., 2018).

Mixed mating may often simply result from accidental fertilisations within and between flowers on the same plant owing to imperfect outcrossing mechanisms (de Jong et al., 1993; Barrett, 2003), but it may also be favoured by selection through several ecological mechanisms (see Goodwillie et al., 2005; Clo et al., 2025 for reviews). These include density- or environment-dependent variation in inbreeding depression (Cheptou and Mathias, 2001; Cheptou and Dieckmann, 2002; Cheptou and Donohue, 2011), limited dispersal of pollen and seed (Ronfort and Couvet, 1995) and uncertain pollination (Morgan and

Wilson, 2005). It is however pollen discounting – the reduction in male mating success through outcrossing that can be caused by traits increasing the selfing rate (Harder and Wilson, 1998) – that has received the most attention. Pollen discounting has been modelled in two ways depending on the mode of selfing considered. In models of prior selfing, where selfing occurs before any opportunity for outcrossing (Lloyd, 1979), pollen discounting is introduced by making individuals' contribution to the outcross pollen pool a function of their selfing rate (e.g., Nagylaki, 1976; Johnston, 1998; Abu Awad and Roze, 2020), reflecting a form of sexual interference whereby floral traits that increase selfing compromise pollen export (Barrett, 2002). By contrast, pollen discounting occurs mechanistically in mass-action models of competing selfing (Lloyd, 1979), where self- and cross-pollen compete for fertilisation. In these models, the selfing rate is determined by the proportion of self-pollen among all pollen deposited on the stigma. Pollen remaining on the plant for selfing is necessarily removed from the contribution individuals make to the outcross pollen pool, resulting in pollen discounting (Holsinger et al., 1984; Holsinger, 1991; Porcher and Lande, 2005). In both prior and competing selfing models, pollen discounting favours intermediate selfing rates because it generates a cost to self-fertilisation that increases non-linearly with the selfing rate (Holsinger, 1991; Johnston, 1998; Porcher and Lande, 2005).

Pollen discounting is therefore thought to be important for selfing evolution, but most of the theory on which this is based assumes fixed sex allocation. All individuals are thus assumed to produce the same number of pollen grains and ovules and so to experience pollen discounting similarly. If sex allocation also evolved, however, the extent of pollen discounting could depend on how much an individual allocates to pollen production. Individuals allocating more heavily to pollen may need to spend a smaller proportion of their pollen to achieve a given selfing rate, so that the relative decrease in their male outcrossing success with the selfing rate (i.e., pollen discounting) might be less severe. Furthermore, empirical evidence suggests that individual selfing rates in fact often vary with individual sex allocation (Damgaard and Abbott, 1995; Chen and Pannell, 2024, 2025).

Selfing and sex allocation might therefore often evolve jointly, yet this joint evolution has rarely been considered. To date, the only model addressing this question is that of Zhang (2000), who studied the joint evolution of sex allocation and pollen export under a mass-action model of competing selfing, where

self- and cross-pollen arrive simultaneously on the stigma and compete for fertilisation (Lloyd, 1979; Holsinger, 1991). Zhang (2000) found that although partial selfing favours female-biased sex allocations (as expected, Charlesworth and Charlesworth, 1981), the equilibrium rate of pollen export favoured by selection is independent of sex allocation, suggesting that the mating system influences sex allocation evolution but is not influenced in return. However, the analyses of Zhang (2000) are limited to directional selection and therefore do not address what happens once the population has converged to the equilibrium. In particular, they do not determine whether this equilibrium remains stable or gives rise to evolutionary branching, whereby distinct types of individuals emerge and become increasingly differentiated, potentially leading to a polymorphic sexual system (Geritz et al., 1998; Pásztor et al., 2016; Avila and Mullan, 2023). Moreover, Zhang (2000)'s model relies on the mass-action assumption, which constrains selfing to be strictly competing and so excludes other modes of self-fertilisation such as prior selfing, where self-pollen fertilises ovules before any opportunity for outcrossing; or delayed selfing, where self-fertilisation only occurs after outcrossing has failed (Lloyd, 1979).

Here, we model the joint evolution of sex allocation and the mating system, allowing for different modes of selfing and investigating what happens beyond directional selection. We introduce an extension of the mass-action model where maternal plants express a trait that allows them to manipulate pollen competition either in favour of or against their own pollen, so that the model encompasses the three modes of selfing identified by Lloyd (1979): prior, competing and delayed. First keeping this trait fixed so that we recover the classical mass-action model (Zhang, 2000), we show that the joint evolution of sex allocation and pollen export can in fact lead to a form of androdioecy where males and partially selfing hermaphrodites coexist. We then let the pollen-competition trait evolve jointly with sex allocation and pollen export, and find that this leads to recurrent episodes of sexual specialisation followed by its collapse, ultimately driving the population to complete selfing through transient androdioecy. Altogether, our results shine light on potential evolutionary feedbacks between plant mating and sexual systems that had so far been overlooked.

2 Model and analytical approach

2.1 Biological scenario

We consider a population composed of a large and constant number N of hermaphroditic plants with an annual life-history, so that generations are non-overlapping. We assume throughout our analyses that plants are haploid with a reduced diploid phase after gamete fusion: ovule fertilisation results in the production of diploid zygotes that rapidly undergo meiosis to give rise to two haploid seeds each. This assumption simplifies the analyses. It allows us to track the evolution of phenotypes in the absence of genetic constraints on the trait combinations that evolve, with in principle no impact on adaptation in the long-term (e.g., Geritz and Kisdi, 2000; Lesaffre et al., 2025; Lehmann and Mullon, 2026). We address the influence of this simplified genetic architecture on our results in the Discussion.

Each individual expresses three continuous traits $\mathbf{z} = (z_1, z_2, z_3)$, each varying between zero and one. The first trait z_1 is sex allocation, defined as the proportion of its reproductive resources that an individual devotes to its female function. The second trait z_2 is the pollen export rate, that is the proportion of its pollen that an individual disperses to fertilise ovules on other plants (while the rest remains on the plant for selfing, as assumed under the mass-action model; Holsinger, 1991). The third trait z_3 , which we refer to as the self-pollen advantage hereafter, allows maternal plants to manipulate the outcome of pollen competition either in favour of or against their own pollen (this will be made explicit in eq. (2a) below). These three traits determine individuals' reproductive behaviour as they pass through the following life cycle events.

(i) *Sexual development*: Plants first grow and mature sexually. Plant $i \in \{1, \dots, N\}$ allocates a fraction $z_{1,i} \in [0, 1]$ of its resources to its female function, and the remaining fraction $1 - z_{1,i}$ to its male function, resulting in the production of

$$F(z_{1,i}) = F_0 z_{1,i} \quad \text{and} \quad M(z_{1,i}) = M_0 (1 - z_{1,i}) \quad (1)$$

ovules and pollen grains, respectively. Throughout, we assume that the number of ovules is large relative

to the number of breeding spots ($N \ll F_0$), and that there is always enough pollen for all ovules to be fertilised ($F_0 \ll M_0$), i.e., no pollen limitation.

(ii) *Mating*: Following pollen and ovule production, individuals export a fraction $z_2 \in [0, 1]$ of their pollen to fertilise ovules on other plants, and the rest $(1 - z_2)$ remains on the plant for self-fertilisation (Holsinger, 1991; Zhang, 2000; Porcher and Lande, 2005). A fraction $c \in [0, 1]$ of the exported pollen is lost during dispersal, and the successful fraction $1 - c$ is distributed equally among the $N - 1$ other plants. The pollen export rate z_2 thus determines the absolute amount of pollen transferred within and between plants. In animal-pollinated species, this rate could be influenced by traits that modulate pollinator attraction (e.g., flower colour or odour), as higher visitation rates should result in higher rates of pollen removal (Johnson and Harder, 2023). In wind-pollinated plants, the pollen export rate might be affected by morphological traits that affect the removal of pollen from anthers by gusts of wind, such as peduncle length or features of pollen morphology (Niklas, 1985).

The self- and cross-pollen received by each plant compete for ovule fertilisation, and this competition is biased by the maternal plant either in favour of or against its own pollen according to its self-pollen advantage $z_{3,i}$. We scale this effect such that competition is unbiased when $z_{3,i} = 1/2$, so that pollen grains from all sources are equally competitive (as envisioned under mass-action), whereas $z_{3,i} > 1/2$ and $z_{3,i} < 1/2$ favour and oppose self-fertilisation, respectively. The self-pollen advantage z_3 thus modulates the relative competitiveness of self- and cross-pollen grains for fertilisation. This trait could be influenced by features of floral morphology that affect the distance between anthers and pistils (herkogamy), or by aspects of floral phenology that affect the timing of anther dehiscence relative to the opening of flowers (dichogamy), both of which may evolve to give either a head-start or a handicap to self-pollen in the race to ovules (Karron et al., 1997; Goodwillie and Weber, 2018). In addition, it could reflect less conspicuous mate choice mechanisms that occur in the style, as maternal plants can hinder (or boost) the progression of pollen tubes from particular sources relative to others in the race towards ovules (Tonnabel et al., 2021; Chenin et al., 2026a,b), as is the case in many self-incompatibility systems for example (Takayama and Akira, 2005; Durand et al., 2020).

Overall, the selfing rate $\alpha(\mathbf{z}_i)$ of individual i expressing phenotype $\mathbf{z}_i = (z_{1,i}, z_{2,i}, z_{3,i})$ when the N

individuals in the population express phenotypes $\mathbf{z}_1, \mathbf{z}_2, \dots, \mathbf{z}_N$ is given by

$$\alpha(\mathbf{z}_i) = \frac{z_{3,i} M_{\text{self}}(\mathbf{z}_i)}{z_{3,i} M_{\text{self}}(\mathbf{z}_i) + (1 - z_{3,i}) \times \underbrace{\frac{1-c}{N-1} \sum_{j \neq i} M_{\text{out}}(\mathbf{z}_j)}_{\text{Total amount of cross-pollen received by plant } i}}, \quad (2a)$$

and the probability that individual i fertilises an ovule of individual $j \neq i$ is

$$\beta(\mathbf{z}_i, \mathbf{z}_j) = \frac{(1 - z_{3,j}) \frac{1-c}{N-1} M_{\text{out}}(\mathbf{z}_i)}{z_{3,j} M_{\text{self}}(\mathbf{z}_j) + (1 - z_{3,j}) \frac{1-c}{N-1} \sum_{k \neq j} M_{\text{out}}(\mathbf{z}_k)}, \quad (2b)$$

where

$$M_{\text{self}}(\mathbf{z}_i) = M(z_{1,i})(1 - z_{2,i})P_{\text{self}} \quad \text{and} \quad M_{\text{out}}(\mathbf{z}_i) = M(z_{1,i})z_{2,i}P_{\text{out}}(z_{3,i}) \quad (2c)$$

respectively give the amount of self-pollen transferred to plant i 's own stigmas and the amount of pollen exported away by plant i . In eq. (2c), $P_{\text{self}} \in [0, 1]$ is a parameter that captures the efficiency of within-plant pollen transfers, while $P_{\text{out}}(z_{3,i}) \in [0, 1]$ captures the efficiency of pollen export, which depends on the self-pollen advantage $z_{3,i}$ of the individual.

We assume that $P_{\text{out}}(z_{3,i})$ decreases with $z_{3,i}$, reflecting the notion that expressing floral features that favour self-pollination might decrease pollen export efficiency, and so lead to pollen discounting due to a form of sexual interference (Barrett, 2002). Mechanistically, this could result from changes to floral morphology that cause self-pollen to be deposited earlier on pistils, and in doing so decrease the efficiency with which pollen is released from anthers and/or exported away from the plant. For example, reduced herkogamy may lead self-pollen to reach the stigma more rapidly, but it may cause the pistil to get in the way of efficient pollen release (Karron et al., 1997; Opedal, 2018). Such a morphological change would give the deposited self-pollen a head-start in the race to ovules, but reduce the amount exported, resulting in pollen discounting.

Since $M_{\text{self}}(\mathbf{z}_i)$ and $M_{\text{out}}(\mathbf{z}_i)$ appear exclusively in ratios in eqs. (2a) and (2b), only their relative scaling

matters. Without loss of generality, we may therefore divide both by P_{self} so that they become

$$M_{\text{self}}(\mathbf{z}_i) = M(z_{1,i})(1 - z_{2,i}) \quad \text{and} \quad M_{\text{out}}(\mathbf{z}_i) = M(z_{1,i})z_{2,i}P(z_{3,i}), \quad (2d)$$

where $P(z_{3,i}) = P_{\text{out}}(z_{3,i})/P_{\text{self}}$ is the relative efficiency of pollen export, which we will work with from now on.

(3) *Seed maturation:* Fertilisation results in the production of diploid zygotes. Outcrossed zygotes become viable with probability one, meanwhile self-fertilised zygotes suffer from inbreeding depression, becoming viable with probability $1 - \delta$, where $\delta \in [0, 1]$ is a parameter measuring the severity of inbreeding depression. Following many authors (e.g., Lande and Schemske, 1985; Lloyd, 1992; Morgan et al., 1997; Rausher and Chang, 1999; Zhang, 2000; Cheptou and Mathias, 2001; Cheptou and Dieckmann, 2002; Tsitroni et al., 2003; Morgan and Wilson, 2005, see Goodwillie et al., 2005 and Clo et al., 2025 for reviews), we therefore assume inbreeding depression to be fixed. We discuss the influence of relaxing this assumption on our results at the end of the paper. Viable diploid zygotes immediately undergo meiosis to give rise to two haploid seeds each.

(4) *Germination and recruitment:* All adults die following seed maturation. The produced seeds then germinate and compete to fill the N breeding spots in the population.

We assume that sex allocation z_1 , the pollen export rate z_2 and the self-pollen advantage z_3 are encoded by a single pleiotropic locus at which alleles may encode any trait combination $\mathbf{z} = (z_1, z_2, z_3) \in [0, 1]^3$ following a continuum-of-alleles model (Kimura, 1965; Walsh and Lynch, 2018). Mutations occur at a small fixed rate at this locus, adding small, unbiased and independent deviations onto the trait values encoded by the mutated copy.

2.2 Invasion analysis

When mutations are rare and have small effects on phenotype, evolution proceeds gradually and its trajectory can be inferred from an invasion analysis (Geritz et al., 1998; Pásztor et al., 2016; Avila and Mullan, 2023). The basis to such analysis is the invasion fitness $W(\mathbf{z}_\bullet, \mathbf{z})$ of a mutant $\mathbf{z}_\bullet = (z_{1\bullet}, z_{2\bullet}, z_{3\bullet})$ ap-

pearing as a single copy in a resident population otherwise fixed for $\mathbf{z} = (z_1, z_2, z_3)$. In a homogeneous population, $W(\mathbf{z}_\bullet, \mathbf{z})$ is given by the number of successfully recruited mutant gene copies produced by a rare focal mutant over a full iteration of the life cycle. In the context of our model, we show in Appendix A.1.1 that invasion fitness can be expressed as

$$W(\mathbf{z}_\bullet, \mathbf{z}) = K(\mathbf{z}) \left[\underbrace{B(\mathbf{z}_\bullet, \mathbf{z})}_{\text{Relative contribution through fertilisation of ovules on other plants}} + \underbrace{\left(1 + \alpha(\mathbf{z}_\bullet)(1 - 2\delta)\right) \frac{F(z_{1\bullet})}{F(z_1)}}_{\text{Relative contribution through own ovules}} \right], \quad (3a)$$

where $K(\mathbf{z}) > 0$ is a positive factor that captures density regulation, and

$$B(\mathbf{z}_\bullet, \mathbf{z}) = \lim_{N \rightarrow \infty} \left(\sum_{i=1}^{N-1} \beta(\mathbf{z}_\bullet, \mathbf{z}) \right) = \frac{(1 - z_3)(1 - c)M_{\text{out}}(\mathbf{z}_\bullet)}{z_3 M_{\text{self}}(\mathbf{z}) + (1 - z_3)(1 - c)M_{\text{out}}(\mathbf{z})} \quad (3b)$$

measures the relative contribution that the mutant makes to the next generation by fertilising ovules on other plants in a large population, i.e., the mutant's total siring success as a pollen donor (see eq. A1 in Appendix A.1.1 for more details).

The term between square brackets in eq. (3a) measures the relative number of mutant copies contributed by the mutant to the next generation, which is composed of two parts. The first part, $B(\mathbf{z}_\bullet, \mathbf{z})$, measures the contribution it makes as a pollen donor (eq. 3b). The second part corresponds to the relative contribution the mutant makes through its own ovules, and can be understood as follows. The mutant produces $F(z_{1\bullet})$ mutant ovules, a proportion $\alpha(\mathbf{z}_\bullet)$ of which get self-fertilised. The resulting self-fertilised zygotes become viable with probability $1 - \delta$, and divide into two haploid mutant seeds when they do so, yielding $2(1 - \delta)$ mutant copies per ovule on average. The remaining ovules $1 - \alpha(\mathbf{z}_\bullet)$ get fertilised by resident pollen, become viable with probability 1 and then divide to give rise to a resident and a mutant seed each (this reduced diploid phase therefore captures the two-fold transmission advantage of selfing, Fisher, 1941). As a result, the average number of mutant copies transmitted per ovule by the mutant through its female function is given by $2(1 - \delta) \times \alpha(\mathbf{z}_\bullet) + 1 \times (1 - \alpha(\mathbf{z}_\bullet)) = 1 + \alpha(\mathbf{z}_\bullet)(1 - 2\delta)$, and the total number of mutant copies contributed by the mutant to the offspring pool is equal to $F(z_{1\bullet}) \times (1 + \alpha(\mathbf{z}_\bullet)(1 - 2\delta))$, which is then scaled by resident female fecundity $F(z_1)$.

Directional selection. Equation (3) can be used to infer the joint evolutionary dynamics of any combination of the three traits, which occur in two phases (Geritz et al., 1998; Leimar, 2009). At first, an initially monomorphic population experiences directional selection, which causes positively selected mutations to rapidly sweep to fixation, so that the population travels through phenotypic space in the direction favoured by selection while remaining largely monomorphic. The direction of selection on a trait z_v , $v \in \{1, \dots, V\}$ (where V denotes the number of jointly evolving traits), is given by the selection gradient, which one can obtain from invasion fitness (eq. 3) as

$$S_v(\mathbf{z}) = \left. \frac{\partial W(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{v\bullet}} \right|_\circ, \quad (4)$$

where the $|_\circ$ notation indicates that the derivative is evaluated at the resident strategy ($\mathbf{z}_\bullet = \mathbf{z}$). A positive selection gradient indicates that directional selection favours an increase in the trait, whereas a negative gradient indicates that it favours a decrease. Through the recurrent fixation of mutations taking the population in the direction favoured by selection, the population may eventually attain a singular strategy $\mathbf{z}^* = (z_1^*, \dots, z_v^*)$ at which directional selection vanishes, i.e.,

$$\mathbf{z}^* \quad \text{such that} \quad \mathbf{S}(\mathbf{z}^*) = \mathbf{0}, \quad (5)$$

where vector $\mathbf{S}(\mathbf{z}) = (S_v(\mathbf{z}))_{1 \leq v \leq V}$ collects the selection gradients on each evolving trait. The mathematical conditions for convergence to a singular strategy are treated in Leimar (2009) (see also Appendix A.1.2.1 here).

Disruptive selection. Once the population has converged to a singular strategy, selection may either be stabilising around $\mathbf{z}^*$, in which case the population remains monomorphic for these trait values and evolutionary dynamics grind to a halt; or selection may be disruptive, potentially leading to evolutionary branching and the gradual emergence of polymorphism (Geritz et al., 1998; Pásztor et al., 2016; Avila and Mullon, 2023). Whether selection is stabilising or disruptive around $\mathbf{z}^*$ is assessed from the Hessian

matrix $\mathbf{H}(\mathbf{z}^*)$, which is a symmetrical $V \times V$ matrix the (u, v) -entry of which is given by

$$H_{uv}(\mathbf{z}^*) = \left. \frac{\partial^2 W(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{u\bullet} \partial z_{v\bullet}} \right|_* \quad (6)$$

for $u, v \in \{1, \dots, V\}$, where the $|_*$ notation indicates that the derivative is evaluated at the singular strategy ($\mathbf{z}_\bullet = \mathbf{z} = \mathbf{z}^*$). Disruptive selection requires the leading eigenvalue of $\mathbf{H}(\mathbf{z}^*)$ to be positive (Leimar, 2009; Débarre et al., 2014; Geritz et al., 2016).

The entries of the Hessian matrix $\mathbf{H}(\mathbf{z}^*)$ carry information regarding selection on each trait and their combination. Each diagonal entry ($u = v$ in eq. 6) gives the quadratic selection coefficients on each trait, informing us on the nature of selection on the trait taken independently (Lande and Arnold, 1983). Specifically, a positive quadratic coefficient $H_{vv}(\mathbf{z}^*) > 0$, $v \in \{1, \dots, V\}$, indicates that selection is disruptive on trait z_v , and a negative coefficient indicates that it is stabilising. The off-diagonal entries ($u \neq v$ in eq. 6), meanwhile, give the correlational selection coefficient between pairs of traits. These tell us whether selection favours a positive ($H_{uv}(\mathbf{z}^*) > 0$) or negative ($H_{uv}(\mathbf{z}^*) < 0$) correlation between two traits within individuals when they evolve jointly (Lande and Arnold, 1983).

A more detailed description of the method is given in Appendix A.1 and an in-depth analysis of the model is available in Appendices A.2 and A.3. We summarise our key findings below.

3 The joint evolution of sex allocation and pollen export enables sexual specialisation

As a starting point, we assume that the self-pollen advantage is fixed at $z_3 = 1/2$, so that pollen competition is unbiased, and that there is no sexual interference ($P(z_3) = 1$). This recovers a classical mass-action model (Holsinger, 1991). In this setting, we analyse the joint evolution of sex allocation z_1 and pollen export z_2 , as previously studied by Zhang (2000), extending their results to consider whether polymorphism can emerge.

3.1 Directional selection leads to outcrossing or mixed mating depending on inbreeding depression

Using eq. (4), we show in Appendix A.2.1 that the selection gradients on sex allocation z_1 and on pollen export z_2 simplify to

$$S_1(\mathbf{z}) = \underbrace{K(\mathbf{z}) \left(1 + (1 - 2\delta)\alpha(\mathbf{z}) \right)}_{>0} \left[\frac{1}{z_1} - (1 - \alpha(\mathbf{z})) \frac{1}{1 - z_1} \right], \quad (7a)$$

$$\begin{aligned} S_2(\mathbf{z}) &= K(\mathbf{z}) \left(\left. \frac{\partial B(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{2\bullet}} \right|_o + (1 - 2\delta) \left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \right|_o \right), \\ &= \underbrace{K(\mathbf{z}) \frac{1 - c}{(1 - cz_2)^2}}_{>0} \left(1 - z_2(1 + c - 2\delta) \right). \end{aligned} \quad (7b)$$

Analysing these gradients jointly reveals that directional selection leads to one of two evolutionary equilibria depending on the severity of inbreeding depression (δ) relative to the costs of pollen dispersal (c), with (z_1, z_2) converging either to

$$\mathbf{z} = \left(\frac{1}{2}, 1 \right), \quad \text{if } \delta \geq \frac{c}{2}, \quad (8a)$$

or to the singular strategy

$$\mathbf{z}^* = (z_1^*, z_2^*) = \left(\frac{1}{2 - \alpha(\mathbf{z}^*)}, \frac{1}{1 + c - 2\delta} \right), \quad \text{if } \delta < \frac{c}{2}. \quad (8b)$$

The first equilibrium corresponds to a population of completely outcrossing hermaphrodites, whereas the second consists of partially selfing, female-biased hermaphrodites, where the implicit expression for z_1^* simplifies to $z_1^* = (1 - 2\delta)/(2 - c - 2\delta)$.

The reason why these two outcomes arise is apparent from the selection gradients. The gradient on sex allocation (eq. 7a) always vanishes at $z_1^* = 1/[2 - \alpha(\mathbf{z}^*)]$, showing that the equilibrium sex allocation is entirely determined by the selfing rate. Under complete outcrossing $\alpha(\mathbf{z}^*) = 0$, selection favours equal investment into male and female function, as predicted by classical theory (Charnov et al., 1976; Charnov, 1982). When selfing occurs ($\alpha(\mathbf{z}^*) > 0$), the equilibrium becomes female-biased because

incomplete pollen dispersal reduces the opportunities for fitness gains through pollen export (see eq. A20a and associated discussion in Appendix A.2.1 for more details). Meanwhile, the selection gradient on pollen export (eq. 7b) is independent of sex allocation z_1 , consistent with previous findings (Zhang, 2000). It reveals a threshold determined by the balance between inbreeding depression δ and dispersal cost c . When $\delta \geq c/2$, selection favours complete pollen export. When $\delta < c/2$, selection instead favours the intermediate pollen export rate $z_2^* = 1/(1 + c - 2\delta)$, which decreases as pollen dispersal becomes more costly and increases as inbreeding depression becomes stronger. Why selection favours an intermediate z_2^* and so mixed mating can be understood by considering a fully selfing population, in which individuals export no pollen ($z_2 = 0$). In such a population, a rare mutant exporting some pollen will maintain the same selfing because it is not receiving any pollen from others, therefore maintaining the same fitness gain as others through its own ovule; while simultaneously gaining reproductive success by fertilising ovules on other plants (who cannot avoid it under strict mass-action). It will thus always be at a fitness advantage.

These results recover those of Zhang (2000) concerning directional selection. In particular, the selection gradient on pollen export is independent of sex allocation. We next investigate whether this independence persists at the level of disruptive selection, or if the joint evolution of sex allocation and pollen export can in fact favour the emergence of polymorphic sexual systems.

3.2 Emergence of androdioecy through evolutionary branching

When $\delta \geq c/2$, eq. (7) shows that pollen export remains under directional selection at the boundary of trait space (i.e., $S_2(1/2, 1) > 0$). The model thus reduces to an outcrossing hermaphroditic population with linear fitness returns through each sexual function. In this situation, equal investment into male and female function, $z_1^* = 1/2$ is under stabilising selection (i.e., $H_{11}(1/2, 1) \leq 0$) and therefore constitutes an evolutionarily stable strategy, as shown previously (Charnov, 1982; Lesaffre et al., 2024, 2025).

When $\delta < c/2$, directional selection leads the population to the mixed mating singular strategy given in eq. (8b), where selection is now determined by the Hessian matrix $\mathbf{H}(\mathbf{z}^*)$ (eq. 6), which here takes the

form of a symmetric 2×2 matrix since only two traits are evolving, with entries

$$H_{11}(\mathbf{z}^*) = \underbrace{K(\mathbf{z}^*)(1 - 2\delta)}_{\text{Positive when } \delta < c/2} \left[\left. \frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet}^2} \right|_* + 2 \left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet}} \right|_* \frac{F'(z_1^*)}{F(z_1^*)} \right], \quad (9a)$$

$$H_{22}(\mathbf{z}^*) = \underbrace{K(\mathbf{z}^*)(1 - 2\delta)}_{>0} \left. \frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}^2} \right|_*, \quad (9b)$$

$$H_{12}(\mathbf{z}^*) = \underbrace{K(\mathbf{z}^*)}_{>0} \left[\left. \frac{\partial^2 B(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{1\bullet} \partial z_{2\bullet}} \right|_* + (1 - 2\delta) \left(\left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \right|_* \frac{F'(z_1^*)}{F(z_1^*)} + \left. \frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet} \partial z_{2\bullet}} \right|_* \right) \right], \quad (9c)$$

where $F'(z_1) = dF(z_1)/dz_1$ denotes the first derivative of female fecundity.

The diagonal elements $H_{11}(\mathbf{z}^*)$ and $H_{22}(\mathbf{z}^*)$ are both negative, indicating that quadratic selection on sex allocation and on pollen export is stabilising when either trait evolves in isolation from the other. For sex allocation (eq. 9a), this is due to two effects. The first is given by the second derivative of the selfing rate (α) with respect to mutant sex allocation, which is negative because the individual's own pollen increasingly competes with itself for self-fertilisation as the amount produced increases, leading to diminishing returns on the selfing rate (analogous to a form of local sperm/mate competition, Hamilton, 1967; Schärer, 2009). The second effect captures the fitness consequences of change in sex allocation through its simultaneous effects on female fecundity (F) and fitness returns per ovule, as influenced by the selfing rate (α). Since any change in sex allocation has opposite effects on these two quantities, this effect is also negative. For pollen export (eq. 9b), the sign of quadratic selection is determined by the second derivative of the selfing rate with respect to pollen export, which is also negative. As the proportion of pollen exported decreases, competition amongst the focal individual's own pollen grains intensifies, again leading to diminishing returns on the selfing rate.

When sex allocation and pollen export evolve jointly, their evolutionary trajectory also depends on correlational selection, which is given by the off-diagonal element $H_{12}(\mathbf{z}^*)$ (eq. 9c). Overall, we find that this element is always negative, indicating that selection favours a negative correlation between the two traits. This result arises because both components of $H_{12}(\mathbf{z}^*)$ favour such a correlation. The first component captures the effect of a joint increase of the two traits on success as a pollen donor. It is always negative because increasing allocation to female function z_1 reduces siring success as a pollen donor, and the effect

of this reduction on fitness is amplified as the focal individual's pollen export increases (intuitively, individuals that gain more fitness through pollen dispersal suffer a bigger fitness cost from reducing pollen production). The second component, which is proportional to $1 - 2\delta$, captures the interaction between z_1 and z_2 on fitness gains through the focal individual's own ovules. Its first part captures the joint effect of a change in pollen export on the selfing rate and a change in sex allocation on relative ovule production. Because increasing pollen export has a negative effect on the selfing rate and increasing sex allocation has a positive effect on ovule production, this part is negative and therefore also favours a negative correlation between pollen export and sex allocation. The second part is the cross-derivative of the selfing rate with respect to both traits. Although it may be either positive or negative depending on parameter values, we find that it is always quantitatively small relative to the other components of eq. (9c) (see Appendix A.2.2 for a detailed discussion). Selection therefore always favours individuals allocating more to their female function to export less pollen, and individuals favouring their male function to export more.

The quadratic selection coefficients in eqs. (9a) and (9b) show that if either trait evolved in isolation from the other, the population would remain monomorphic under stabilising selection. When the two traits co-evolve, however, correlational selection can generate disruptive selection if it is strong relative to quadratic selection on both traits, specifically if $H_{12}(\mathbf{z}^*)^2 > H_{11}(\mathbf{z}^*)H_{22}(\mathbf{z}^*)$. We show in Appendix A.2.2 that this condition is always satisfied, indicating that the Hessian matrix always has a positive eigenvalue (Fig. 1A) and selection favours the emergence of polymorphism through evolutionary branching. Intuitively, this result can be understood by keeping in mind that inbreeding depression is weak here ($\delta < c/2$), so that self-fertilised seeds provide greater average fitness returns than outcrossed seeds. In this setting, an individual allocating more resources to ovule production is also able to increase the fitness returns it gains per ovule by simultaneously reducing its pollen export rate. This creates a multiplicative increase in fitness gains through female function that favours specialisation into the production of self-fertilised seeds. However, as individuals cannot avoid fertilisation by others under strict mass-action, the emergence of such a female-biased type also creates more opportunities for fitness gains as a pollen donor, favouring individuals that simultaneously allocate more resources to their male function and export pollen at a higher rate.

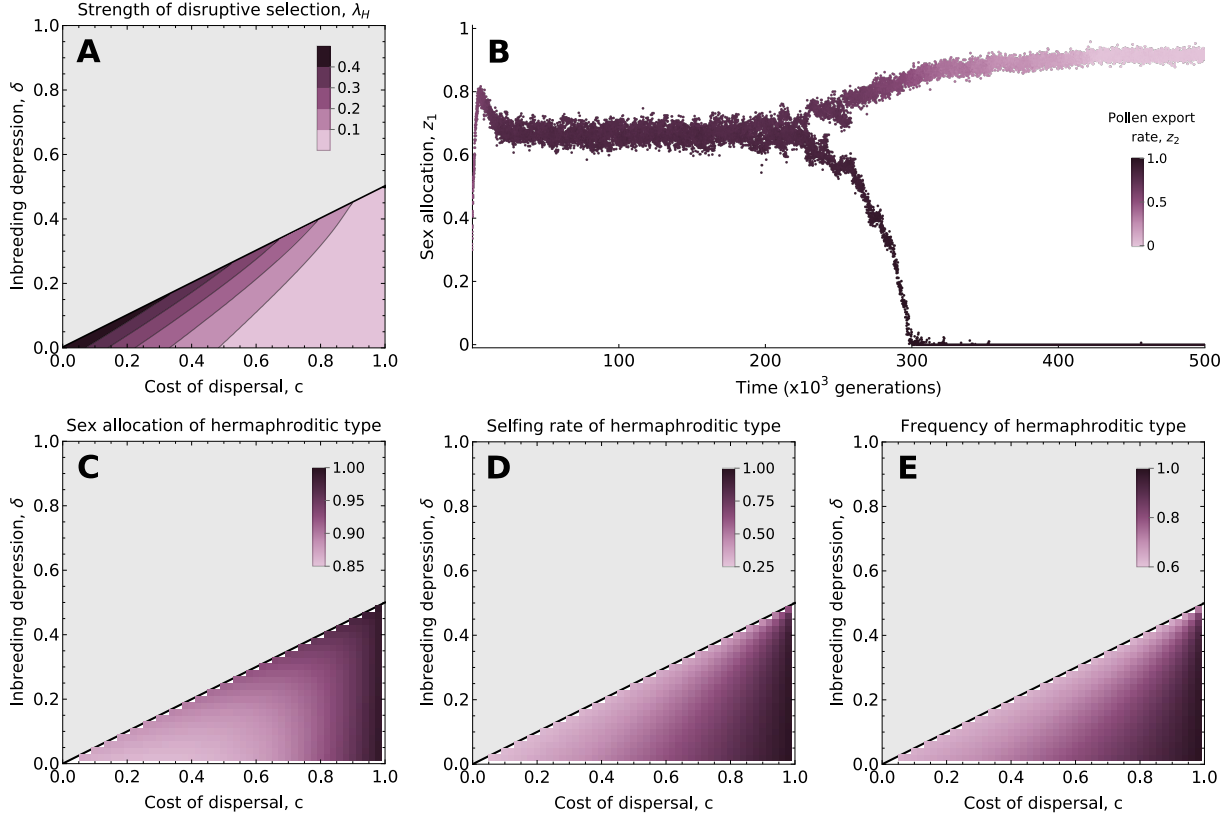


Figure 1: Joint evolution of sex allocation and pollen export produces stable androdioecy. **A** Strength of disruptive selection as given by the leading eigenvalue of the Hessian matrix as a function of the cost of pollen dispersal (c) and inbreeding depression (δ). Values are only represented where $\delta < c/2$, as the population evolves complete outcrossing and remains monomorphic otherwise. Darker colours indicate stronger disruptive selection. **B** Evolutionary trajectory in an individual-based simulation. Each point gives the sex allocation (y-axis) of a randomly sampled individual in the population at a particular time (x-axis). The purple shade gives the pollen export rate of that individual, with darker colours indicating higher values. We see that the population first converges to an intermediate sex allocation and pollen export strategy, and then gradually splits into two types: a female-biased hermaphroditic type that does not export any pollen (lighter branch) and a pure male type that exports all its pollen (darker branch). Parameter values used: $c = 0.75$, $\delta = 0.25$, $N = 10^4$, $\mu = 10^{-3}$, $\sigma = 10^{-2}$. **C-D-E** Predicted sex allocation (**C**), selfing rate (**D**) and frequency (**E**), of the hermaphroditic type at evolutionary equilibrium, as a function of the cost of pollen dispersal (c) and inbreeding depression (δ).

We ran numerical analyses as well as individual-based simulations to investigate the nature of polymorphism evolving after evolutionary branching (Appendix A.1.3.2 for details). These analyses reveal that evolutionary dynamics lead to the coexistence of two highly differentiated types (Fig. 1B): a male type that produces only pollen and exports all of it (i.e., expressing $z_1 = 0$ and $z_2 = 1$), and a hermaphroditic type that specialises into ovule production and self-fertilisation, producing little pollen and exporting none of it ($z_1 > 1/2$ and $z_2 = 0$). The hermaphroditic type is heavily female-biased (high z_1 , Fig. 1C) and self-fertilises at a rate that varies depending on how much pollen it receives from males, from about 0.25 when the cost of pollen dispersal is low or inbreeding depression is close to $c/2$, to almost 1 when the cost of dispersal is high and inbreeding depression is low (Fig. 1D). This type is always the most frequent in the population, which is due to the asymmetry in transmission created by selfing: males can only ever produce one male offspring per ovule fertilised on a hermaphrodite, whereas hermaphrodites not only produce one hermaphroditic offspring per outcrossed ovule, but also produce two per selfed ovule (Fig. 1E). Thus, the joint evolution of sex allocation and pollen export leads to the emergence of a form of androdioecy where a male type exporting large amounts of pollen persists in the population by fertilising a fraction of the ovules produced by a hermaphroditic type, which is seeking to self-fertilise as many of its ovules as possible. This polymorphism is evolutionarily stable because pollen from different sources compete on equal grounds for fertilisation under classical mass-action, so that hermaphrodites cannot prevent being fertilised by males. We relax this assumption in the next section.

4 Maternal control of pollen competition and the rise and fall of sexual specialisation

We now allow maternal plants to manipulate the outcome of pollen competition. Specifically, we let the self-pollen advantage z_3 evolve jointly with sex allocation (z_1) and pollen export (z_2). The resulting

selection gradients, derived in Appendix A.3, are given by

$$S_1(\mathbf{z}) = K(\mathbf{z}) \left(1 + (1 - 2\delta)\alpha(\mathbf{z}) \right) \left[\frac{1}{z_1} - (1 - \alpha(\mathbf{z})) \frac{1}{1 - z_1} \right], \quad (10a)$$

$$S_2(\mathbf{z}) = K(\mathbf{z}) \frac{\alpha(\mathbf{z})(1 - \alpha(\mathbf{z}))}{z_2(1 - z_2)} \left[1 - z_2 \left(2(1 - \delta) - P(z_3)(1 - c) \frac{1 - z_3}{z_3} \right) \right], \quad (10b)$$

$$S_3(\mathbf{z}) = K(\mathbf{z}) \frac{\alpha(\mathbf{z})(1 - \alpha(\mathbf{z}))}{z_3(1 - z_3)} \left[1 - 2\delta + (1 - z_3) \left(z_3 + \frac{z_2(1 - c)}{1 - z_2} (1 - z_3) P(z_3) \right) \frac{P'(z_3)}{P(z_3)} \right], \quad (10c)$$

where $P'(z_3)$ denotes the first derivative of the pollen export efficiency function $P(z_3)$.

Eq. (10a) shows that directional selection on sex allocation z_1 remains unchanged when z_3 evolves jointly with the other traits (compare this equation with eq. (7a) in the previous section), so the singular sex allocation strategy is still given by the one shown in eq. (8b). By contrast, eqs. (10b) and (10c) show that selection gradients on pollen export rate z_2 and the self-pollen advantage z_3 depend on each other in a complicated way. To understand how selection shapes the joint evolution of these two traits, we first consider the simpler case in which self-pollen promotion does not cause pollen discounting ($P(z_3) = 1$ for all z_3).

4.1 Cost-free manipulation of pollen competition leads to pure selfing or outcrossing

Assuming $P(z_3) = 1$ for all z_3 , the selection gradients on pollen export and self-pollen advantage reduce to

$$S_2(\mathbf{z}) = K(\mathbf{z}) \frac{\alpha(\mathbf{z})(1 - \alpha(\mathbf{z}))}{z_2(1 - z_2)} \left[1 - z_2 \left(2(1 - \delta) - (1 - c) \frac{1 - z_3}{z_3} \right) \right], \quad (11a)$$

$$S_3(\mathbf{z}) = K(\mathbf{z}) \frac{\alpha(\mathbf{z})(1 - \alpha(\mathbf{z}))}{z_3(1 - z_3)} (1 - 2\delta). \quad (11b)$$

The joint analysis of these two gradients reveals two possible outcomes of directional selection that are determined by the sign of selection on the self-pollen advantage (eq. 11b): either (i) inbreeding depression is strong relative to the transmission advantage ($\delta > 1/2$) and the population evolves to a complete outcrossing equilibrium where $z_3 = 0$ (self-pollen cannot fertilise ovules), at which point selection favours

complete pollen export ($z_2 = 1$), and equal allocation male and female function ($z_1 = 1/2$); or (ii) inbreeding depression is weak ($\delta < 1/2$) and the population evolves towards a complete selfing equilibrium where $z_3 = 1$ (so that only self-pollen can fertilise ovules), in which case pollen export z_2 becomes effectively neutral as no fitness can be gained as a pollen donor and negligible pollen is needed for selfing (i.e., $S_2(\mathbf{z}) = 0$ when $z_3 = 1$ so that any $z_2 < 1$ is stable, see Appendix A.3.1 for details), and selection favours a female-biased sex allocation as given by eq. (8b). Thus, there can be no mixed mating equilibrium when maternal plants can manipulate pollen competition with no consequence for their pollen export, as predicted by classical theory on prior selfing evolution in the absence of pollen discounting (Lande and Schemske, 1985).

4.2 Pollen discounting can generate a mixed mating equilibrium

Eq. (10c) shows that pollen discounting ($P'(z_3) < 0$) generates a strictly negative term that selects against self-fertilisation. This countervailing effect could favour the existence of an intermediate equilibrium self-pollen advantage and hence mixed mating. We show in Appendix A.3.2.1 that complete outcrossing ($z_2 = 1$ and $z_3 = 0$) and complete selfing ($z_2 < 1$ and $z_3 = 1$) remain possible endpoints of gradual evolution with pollen discounting. Because eq. (10) does not yield explicit predictions for the outcome of directional selection, we analysed it numerically after specifying the pollen export efficiency function $P(z_3)$, which we assume to be given by

$$P(z_3) = \frac{2(1 - e^{\kappa/2})}{2 - e^{\kappa/2} - e^{\kappa z_3}} \quad (12)$$

(see Fig. 2A). Equation (12) says $P(z_3)$ is a decreasing sigmoid function with z_3 that satisfies $P(1/2) = 1$, so that our previous model without maternal control on pollen competition is recovered for $z_3 = 1/2$. The shape of $P(z_3)$ is controlled by parameter $\kappa > 0$. A higher κ means stronger sexual interference and so more severe pollen discounting (Fig. 2A).

In addition to the complete selfing and complete outcrossing equilibria, pollen discounting can generate an equilibrium at which the population shows mixed mating (Fig. 2B). The existence of such an equilibrium

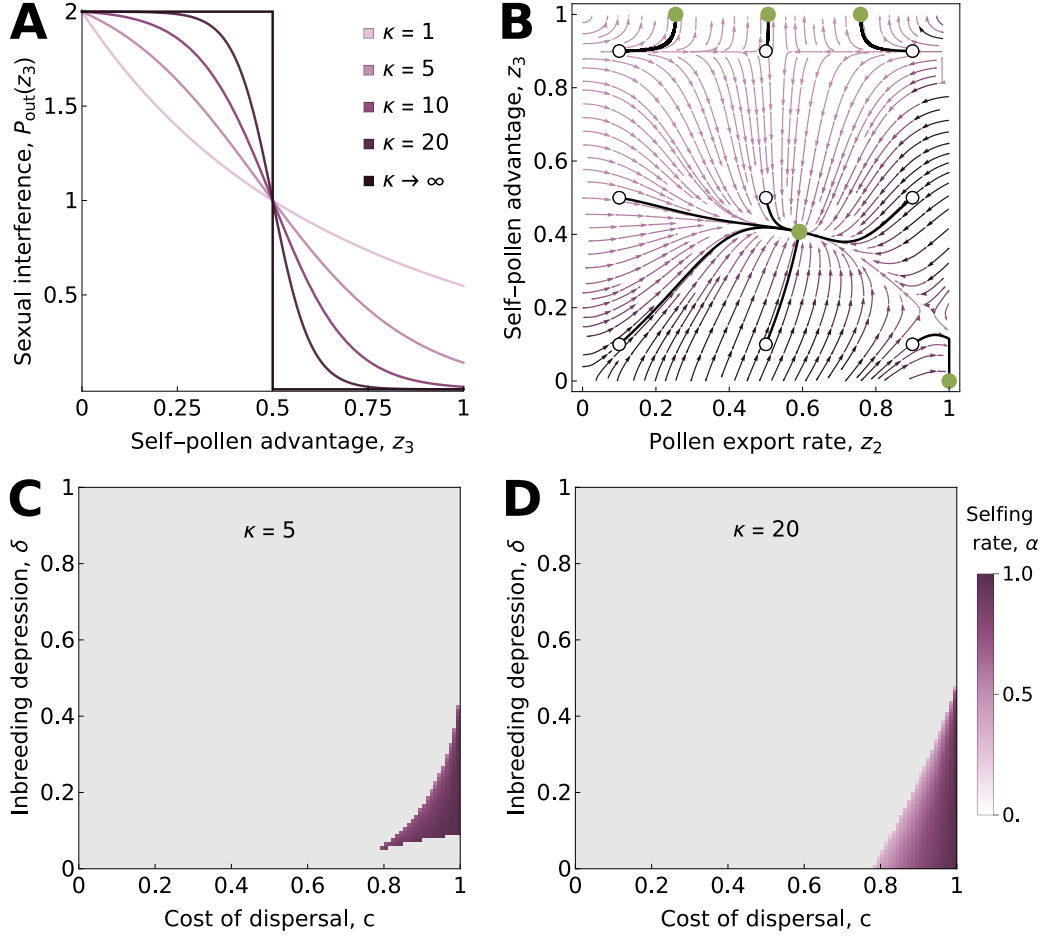


Figure 2: Pollen discounting can generate a mixed-mating equilibrium under maternal control of pollen competition. **A** Pollen export efficiency $P(z_3)$ as a function of z_3 for different values of κ . Darker shades of purple indicate higher values of κ and so stronger sexual interference. **B** Deterministic evolutionary trajectories of pollen export rate z_2 and self-pollen advantage z_3 under directional selection, as inferred from a numerical analysis. The arrows indicate the direction of selection for a given point in the (z_2, z_3) plane, and their colour indicates the intensity of selection with darker shades indicating stronger selection. White points correspond to the nine starting points used in our numerical analyses, solid black lines show the inferred trajectories from these starting points and green points show the attained equilibria. Trajectories reach one of three equilibria: complete outcrossing (bottom right corner), mixed mating (intermediate point) and complete selfing (top edge). The algorithm stops for different values of z_2 in the complete selfing case because the selection gradient on pollen export is zero once self-pollen advantage is $z_3 = 1$. Parameters used: $\kappa = 10$, $c = 0.9$, $\delta = 0.05$. **C-D** Selfing rate at the mixed mating equilibrium, where it exists, for moderate ($\kappa = 5$) and strong sexual interference ($\kappa = 20$) as a function of the cost of pollen dispersal c and inbreeding depression δ . Coloured tiles indicate the selfing rate at the mixed mating equilibrium, and the grey area indicates the parameter range in which no such equilibrium was found.

requires a high cost of pollen dispersal and substantial pollen discounting (high c and κ). Where it exists, the selfing rate increases with the cost of pollen dispersal c and decreases with inbreeding depression δ . Stronger pollen discounting produces a broader range of equilibrium selfing rates (Fig. 2C-D). At this mixed mating equilibrium, the population expresses a female-biased sex allocation given by eq. (8b).

4.3 Evolutionary branching leads to complete selfing through transient androdioecy

As shown in section 4.2, directional selection may lead the population to three possible equilibria under pollen discounting: complete selfing, complete outcrossing, or a mixed-mating equilibrium, depending on parameters and initial conditions. Complete selfing and complete outcrossing are endpoints of gradual evolution (see Appendix A.3.2.1 for details), but at the mixed-mating equilibrium, second-order effects of selection may potentially lead to evolutionary branching (since the selection gradient on each trait is zero at this equilibrium).

To study this, we computed $\mathbf{H}(\mathbf{z}^*)$ numerically at the mixed mating equilibrium for a wide range of parameter combinations. For all of these, we found that $\mathbf{H}(\mathbf{z}^*)$ has a positive eigenvalue, indicating that selection is disruptive and favours evolutionary branching. Further analysis of $\mathbf{H}(\mathbf{z}^*)$ reveals that evolutionary branching favours the same kind of polymorphism as before: it favours the emergence of a hermaphroditic type specialising in the production of self-fertilised ovules, and a male type specialising in pollen export. Here, however, the types also diverge in self-pollen advantage: the hermaphroditic type evolves a higher value, which increases its selfing rate, whereas the male evolves a lower value, which improves its pollen export. As the hermaphroditic type evolves an increasingly high self-pollen advantage, fertilisation of its ovules by the male type becomes increasingly unlikely. Male reproductive success and frequency therefore decline until the male type goes extinct (Fig. 3A).

Once the male type disappears, the population becomes monomorphic for the hermaphroditic type and therefore again evolves under directional selection. Its subsequent trajectory depends on the hermaphroditic phenotype when the male type disappears. If the male type goes extinct when the self-pollen advantage of the hermaphroditic type lies outside the basin of attraction of the complete selfing equilibrium (i.e., sufficiently far from $z_3^* = 1$; below the dotted line in Fig. 3B), then the population is driven back to the mixed mating equilibrium, at which point disruptive selection again leads to evolutionary branching. If the hermaphroditic type has evolved a high enough self-pollen advantage by the time males disappear, on the other hand, the population evolves towards complete selfing ($z_3^* = 1$) and gradual evolutionary dynamics grind to a halt.

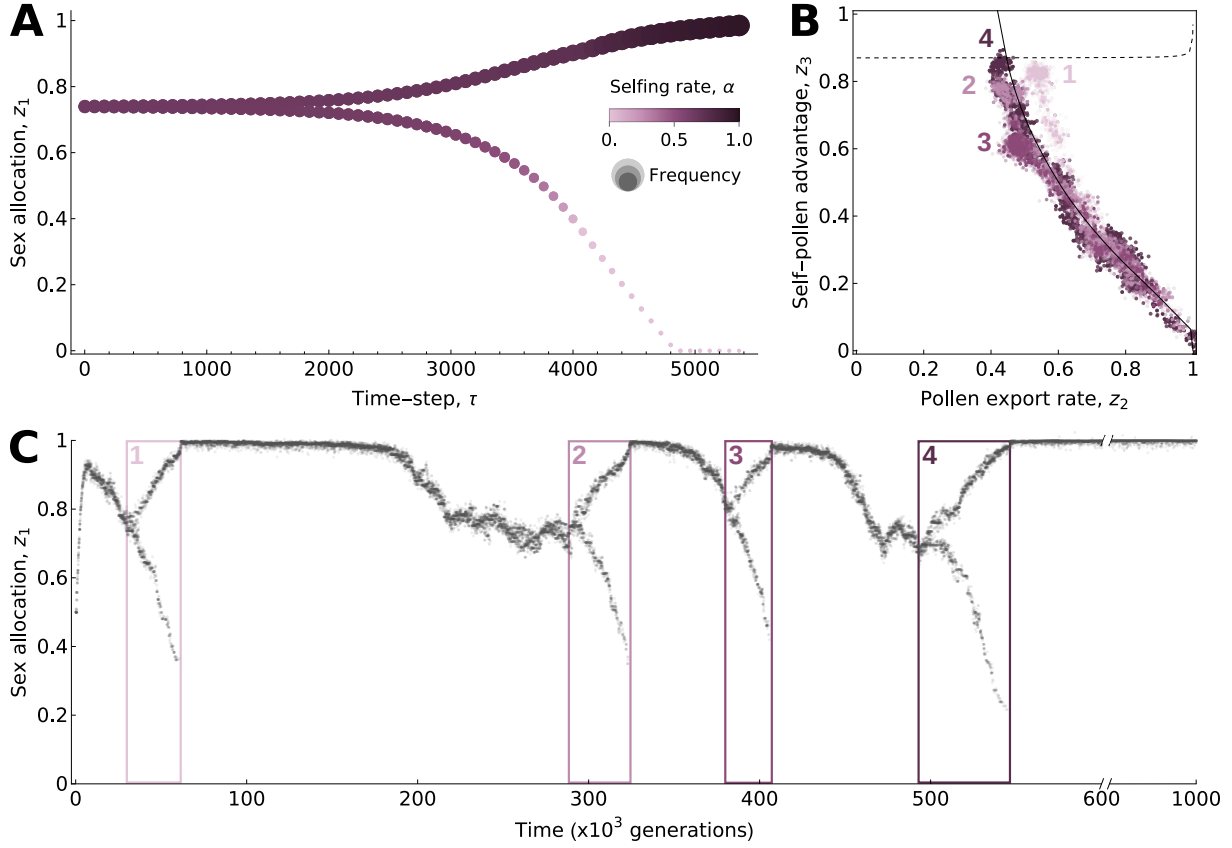


Figure 3: Evolutionary branching produces transient androdioecy and ultimately complete selfing. **A** Deterministic evolutionary trajectory of sex allocation following branching, as inferred by a numerical analysis. Point size indicates the frequency of the type and the shade of purple indicates its selfing rate. As the two types diverge, the selfing rate of the hermaphroditic type increases and the frequency of the male type decreases. **B** Evolutionary trajectory in the (z_2, z_3) phenotypic plane. The solid line gives the predicted trajectory from the numerical analysis (as shown in **A**) and the dotted line gives the threshold self-pollen advantage value above which directional selection favours complete selfing in a monomorphic population. If the male type goes extinct when the hermaphroditic type is below this line, the population evolves back to the mixed mating equilibrium and undergoes evolutionary branching again. Otherwise, the population evolves towards complete selfing. The points show the trajectories followed by the four branching events that took place until complete selfing evolved in the individual-based simulation presented in **C**. The points are coloured per branching event. **C** An example of a simulation where the population undergoes evolutionary branching and evolves back to the mixed equilibrium three times before eventually evolving complete selfing. Parameter set used in all plots: $\kappa = 5$, $c = 0.95$, $\delta = 0.25$. Other parameters for simulations: $N = 5 \times 10^3$, $\mu = 10^{-3}$, $\sigma = 10^{-2}$.

In stochastic individual-based simulations, both outcomes may be observed with probabilities that depend on the parameter values, which determine the strength of disruptive selection and the rate at which the two types diverge, and on population size, which determines how long the male type is expected to persist at low frequencies. However, because complete selfing is an evolutionary endpoint (whereas mixed mating never is in this model), repeated cycles of evolutionary branching and male extinction eventually drive the population to complete selfing (Fig. 3B-C).

5 Discussion

Selfing strongly influences sex allocation evolution, yet most theoretical studies treat the mating system as fixed. Models of selfing evolution likewise seldom consider the evolution of sex allocation. In the only study to address their joint evolution, Zhang (2000) modelled the joint evolution of sex allocation and pollen export under a mass-action model of self-fertilisation (Holsinger, 1991). They showed that directional selection on the pollen export rate should be independent of sex allocation, suggesting that selfing influences the evolution of sex allocation but is not influenced in return. Here, we show that selfing and sex allocation do in fact influence the evolution of one another: specifically, correlational selection favours associations between sex allocation and mating-system traits that can generate polymorphic sexual systems.

Under the classical mass-action model, the joint evolution of sex allocation and pollen export leads to a form of androdioecy where hermaphrodites that specialise in the production of self-fertilised ovules coexist with males that specialise into pollen export. To our knowledge, androdioecy where hermaphroditic individuals do not export any male gamete has not been observed in any plant species. On the other hand, it appears to be more common in animals, notably in nematodes (e.g., in *Caenorhabditis elegans*, Denver et al., 2011) and branchiopod crustaceans (Weeks, 2012), where males coexist with hermaphroditic individuals that do not transfer any sperm (see Chap. 1 in Leonard, 2018 for a review). Our findings thus reveal a possible path from simultaneous hermaphroditism to androdioecy through the joint evolution of sexual and mating strategies that could help explain the evolution of this rare sexual system in some of these animals. However, in most known cases of androdioecy in animals, phylogenetic evidence indicates

that this sexual systems evolved from dioecy, probably in response to selection for reproductive assurance in females (Denver et al., 2011; Weeks, 2012; Leonard, 2018). Furthermore, the androdioecy that emerges from the joint evolution of sex allocation and pollen export here is only stable because hermaphrodites cannot avoid fertilisation by males. Androdioecy is never stable once this constraint is lifted, as we showed in our extended model.

When individuals can manipulate gamete competition in favour of their own, we found that the joint evolution of pollen export and the self-pollen advantage makes the existence of a mixed mating equilibrium less likely under directional selection, because prior selfing is now able to evolve where constraints prevented it to do so under classical mass-action. Where a mixed mating equilibrium is reached, evolutionary branching again occurs, leading to the emergence of self-fertilising hermaphrodites and males. However, this polymorphism is no longer stable because hermaphrodites progressively exclude male pollen by evolving a higher self-pollen advantage, causing the extinction of males and thereby driving the evolution of complete selfing through the emergence and collapse of androdioecy. These results show that, contrary to when selfing evolves in isolation (Johnston, 1998; Porcher and Lande, 2005), pollen discounting is not sufficient to ensure the long-term maintenance of mixed mating when sex allocation evolves as well. Moreover, these results expose a limitation of the mass-action model for explaining the widespread occurrence of mixed mating, despite its having sometimes been used for this purpose (e.g., Holsinger, 1991; Porcher and Lande, 2005). Under mass-action, individuals cannot avoid fertilisation by pollen from other individuals, so some outcrossing is unavoidable whenever pollen is exchanged. Mixed mating is therefore built into the model rather than maintained by selection. This highlights the importance of including prior selfing in models seeking to explain the maintenance of mixed mating, especially since this form of selfing is straightforward to achieve mechanistically (Goodwillie et al., 2005).

In our model, mixed mating is unstable because correlated changes in sex allocation and mating-system traits produce increasing fitness returns that favour specialisation and evolutionary branching. Ecological processes that generate diminishing returns to sexual specialisation could oppose this effect. A likely candidate is limited dispersal of pollen and seeds (Fromhage and Kokko, 2010). Limited seed dispersal causes related seeds to compete with one another for recruitment. As investment in female function rises, com-

petition among a plant's related seeds intensifies, reducing expected fitness returns per seed and hence the benefit of female specialisation ("local resource competition"; Chap. 8 in Maynard Smith, 1978, Clark, 1978; Charnov, 1982; Lloyd and Bawa, 1984; Fromhage and Kokko, 2010; Biernaskie, 2010; Lesaffre et al., 2025). Limited pollen dispersal leads to a similar effect on fitness returns per pollen grain and therefore selects against male specialisation ("local mate/sperm competition"; Hamilton, 1967; Schärer, 2009; Fromhage and Kokko, 2010; Lesaffre et al., 2025). Together, limited dispersal of pollen and seeds could stabilise mixed mating by opposing evolutionary branching in sex allocation. Thus, mechanisms that maintain hermaphroditism may also stabilise mixed mating. Beyond its effect on branching and sex allocation, limited dispersal may favour intermediate selfing rates on its own (Ronfort and Couvet, 1995), but the underlying mechanisms require further investigation.

We assumed that all traits are encoded by a single pleiotropic locus and that individuals are haploid, thereby eliminating potential genetic constraints on phenotypic evolution. Such constraints may oppose evolutionary branching by hindering the build-up of the phenotypic associations favoured by correlational selection, and could thus prevent the emergence of self-fertilising hermaphrodites and males. However, although genetic constraints may pose a strong barrier to the emergence of polymorphism in the short-term, selection may resolve them in the long-term. Selection favouring phenotypic polymorphism can itself reshape genetic architecture and build the trait associations favoured by correlational selection (Rueffler et al., 2006; van Doorn and Dieckmann, 2006; Kopp and Hermisson, 2006; Yeaman and Whitlock, 2011), a process explored in detail by Lesaffre et al. (2025) in the case of the joint evolution of sex allocation and a trait with opposing effects on male and female components of individual fitness. In the biological scenario investigated here, once a mixed-mating equilibrium exists, selection favours evolutionary branching when inbreeding depression is sufficiently weak relative to the cost of pollen dispersal, which is also a condition for the equilibrium to exist. Correlational selection promoting evolutionary branching is therefore expected to persist for as long as the conditions for a mixed mating equilibrium are met, thereby giving the genetic architecture time to evolve.

In the mating model proposed here, the pollen export rate z_2 determines the absolute amount of self-pollen that remains on the plant for selfing, and the self-pollen advantage z_3 determines whether self-

pollen grains are at a (dis)advantage relative to cross-pollen grains in the race to ovules. We assumed throughout that these two traits were the target of independent mutations, so that any correlation between them was free to evolve under the action of selection. Biologically, however, changes to floral morphology might sometimes affect both traits simultaneously. For example, reducing herkogamy could increase the absolute amount of self-pollen deposited on a plant's stigma (decreasing z_2) in addition to causing it to gain a head-start by being deposited earlier (increasing z_3). In this case, mutations modifying the degree of herkogamy would affect both z_2 and z_3 and generate a negative correlation between the two traits due to mutational constraints. Such a negative correlation would accelerate the evolutionary dynamics described here, as it aligns with the direction of correlation selection. Conversely, a positive correlation would slow down the response to selection (Arnold et al., 2008). This reinforces the notion that valuable insights into the joint evolution of sexual and mating systems could be gained from incorporating the details of floral morphology into theoretical models.

Treating inbreeding depression as fixed is conservative with respect to our conclusion that mixed mating is evolutionarily unstable. The mutation load that underlies inbreeding depression is expected to evolve with the selfing rate, but the main effect of selfing on the load is to improve purging, causing inbreeding depression – and so the cost of selfing – to decrease as the selfing rate increases (Lande and Schemske, 1985; Charlesworth et al., 1990, 1991; Abu Awad and Roze, 2020). Purging therefore tends to make the maintenance of mixed mating less likely because it creates a positive feedback loop towards complete selfing, which means that assuming a fixed level of inbreeding depression gives mixed mating a better shot at persisting. More importantly, the correlational selection effect that we describe between sex allocation and mating-system traits does not rely on this assumption: it only requires inbreeding depression to be weak enough for selfing to be net beneficial to maternal parents. We therefore expect that accounting for the dynamics of the load would not fundamentally alter the insights obtained here. Interestingly, recent empirical evidence in hermaphroditic snails suggests that recessive deleterious mutations may not be the only source of inbreeding depression, as epigenetic marks were shown to generate a sizeable reduction in inbred fitness (Bonel et al., 2026). These marks accumulate rapidly and could therefore form the basis of a component of inbreeding depression that is less readily eliminated by purging.

In summary, our results show that the mating and sexual systems of flowering plants may often evolve jointly. We found that this joint evolution can lead to evolutionarily stable androdioecy when plants cannot avoid fertilisation by other individuals. When maternal plants can manipulate the outcome of pollen competition, the joint evolution of selfing and sex allocation ultimately leads to complete selfing through the rise and fall of sexual specialisation. These processes require inbreeding depression to be weak relative to the transmission advantage of selfing. In the future, investigating how additional features of plant ecology, such as pollen limitation or limited dispersal, influence this joint evolution may yield valuable insight into the diversity of plant reproductive strategies.

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Appendices to

The evolutionary interplay between self-fertilisation and sex allocation in flowering plants

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Appendix A

Mathematical and numerical analyses

In this Appendix, we introduce the life cycle followed by individuals in the model and the methodology we use to analyse it. We then provide a detailed account of our analyses.

A.1 Methods

A.1.1 Life cycle and invasion fitness

Here, we introduce the life cycle that individuals follow. It is presented in a way that does not make any assumption on how traits affect gamete production or mating success through selfing and outcrossing, which allows us to gain general insight into the pathways through which selection acts in this model, irrespective of the particular trait.

We consider a large population composed of N haploid hermaphrodites with discrete generations, where individuals are characterised by their phenotype $\mathbf{z}_i \in \mathbb{R}^V$ for individual $i \in \{1, \dots, N\}$, where V is the number of evolving traits. Each generation, individuals pass through the following life-cycle events.

(i) *Sexual development*: First, plants grow and mature sexually, allocating resources to their female and male function according to their phenotype, resulting in the production of $F(\mathbf{z}_i)$ ovules and $M(\mathbf{z}_i)$ pollen grains by plant i . Throughout, we assume that the number of ovules is large relative to the number of breeding spots available in the population, so that there always is enough seeds produced to fill these spots and the population remains of constant size N . Further, we assume that pollen is much less costly to produce than ovules so that there is always enough pollen for all produced ovules to be fertilised (no pollen limitation).

(ii) *Mating*: Following pollen and ovule production, plants disperse a fraction of their pollen to fertilise

ovules on other plants. Pollen grains received by each plant then compete to fertilise the ovules it carries. The pollen dispersal strategy of individual i and how pollen competition plays out in its pistils depends on its phenotype $\mathbf{z}_i$ as well as the phenotypes of all other individuals in the population, so that individual i self-fertilises at a rate $\alpha(\mathbf{z}_i)$ and the probability that it fertilises an ovule of individual $j \neq i$ is given by $\beta(\mathbf{z}_i, \mathbf{z}_j)$.

(3) *Seed maturation:* Fertilisation results in the production of diploid zygotes. Self-fertilised zygotes suffer from inbreeding depression, becoming viable with probability $1 - \delta$, where $\delta \in [0, 1]$ measures the magnitude of inbreeding depression; meanwhile outcrossed zygotes become viable with probability one. Viable diploid zygotes immediately undergo meiosis to give rise to two haploid seeds each.

(4) *Germination and recruitment:* Following seed maturation, all adults die and the produced seeds germinate and compete for recruitment to fill the N breeding spots in the population.

We assume that individual phenotype $\mathbf{z}_i \in \mathbb{R}^V$ is encoded by a single pleiotropic locus at which alleles may encode any trait combination (or equivalently, V fully linked loci each encoding one trait; “continuum-of-alleles” model, Kimura, 1965; Walsh and Lynch, 2018). Mutations occur at a small fixed rate at this locus, adding small, unbiased and independent deviations onto the trait values encoded by the mutated copy. Under these assumptions, evolution proceeds gradually and its dynamics can be inferred using an adaptive dynamics approach (Avila and Mullan, 2023).

This approach leverages population genetics results to infer the long-term evolutionary trajectory of a population for traits subject to rare mutations with small, unbiased phenotypic effects. The fundamental basis of the adaptive dynamics approach is the *invasion fitness* of a mutant encoding phenotype $\mathbf{z}_\bullet \in \mathbb{R}^V$ arising as a single copy in a resident population fixed for $\mathbf{z} \in \mathbb{R}^V$ at its ecological equilibrium, $W(\mathbf{z}_\bullet, \mathbf{z})$. In a homogeneous population, invasion fitness can be obtained by computing the number of mutant gene copies contributed by a rare focal mutant to the next generation. Under the life cycle presented above, the

invasion fitness of a mutant $\mathbf{z}_\bullet$ arising as a single copy in a population otherwise fixed for $\mathbf{z}$ is given by

$$W(\mathbf{z}_\bullet, \mathbf{z}) = \left[\underbrace{F(\mathbf{z}_\bullet) \left(2\alpha(\mathbf{z}_\bullet)(1 - \delta) + 1 - \alpha(\mathbf{z}_\bullet) \right)}_{\text{Number of mutant seeds produced from own ovules}} + \underbrace{\sum_{i=1}^{N-1} F(\mathbf{z}) \beta(\mathbf{z}_\bullet, \mathbf{z})}_{\text{Number of mutant seeds produced by fertilising resident ovules}} \right] \times k(\mathbf{z}_\bullet, \mathbf{z}), \quad (\text{A1a})$$

where

$$k(\mathbf{z}_\bullet, \mathbf{z}) = \frac{N}{2(N-1)F(\mathbf{z})[1 - \delta\alpha(\mathbf{z})] + 2F(\mathbf{z}_\bullet)[1 - \delta\alpha(\mathbf{z}_\bullet)]} \quad (\text{A1b})$$

denotes the probability that a given seed is recruited in the population.

Equation (A1) can be understood as follows. The first term between square brackets gives the number of mutant seeds produced by the mutant through its own ovules. The mutant produces $F(\mathbf{z}_\bullet)$ ovules carrying $\mathbf{z}_\bullet$. These ovules then get self-fertilised in proportion $\alpha(\mathbf{z}_\bullet)$, resulting in selfed zygotes that become viable with probability $1 - \delta$ and divide into two mutant haploid seeds if they do so. The rest of the ovules $(1 - \alpha(\mathbf{z}_\bullet))$ gets fertilised by resident pollen, resulting in outcrossed, heterozygous zygotes that become viable with probability 1 and subsequently divide to produce one mutant seed and one resident seed. This reduced diploid phase therefore captures the transmission advantage of self-fertilisation, which is diminished by inbreeding depression (Fisher, 1941; Lande and Schmske, 1985). The second term between square brackets gives the number of mutant seeds that the mutant produces by fertilising ovules on resident individuals. Each of the $N - 1$ resident individuals in the population produces $F(\mathbf{z})$ ovules, a fraction $\beta(\mathbf{z}_\bullet, \mathbf{z})$ of which gets fertilised by the mutant. Thus, the term between square brackets on the first line of eq. (A1) gives the number of mutant seeds produced by the focal mutant through all reproductive pathways. The quantity $k(\mathbf{z}_\bullet, \mathbf{z})$ gives the probability that a seed gets recruited in the population in the next time-step, which is obtained as the number of available breeding spots (N) divided by the total number of seeds produced by the population.

In the limit where the population becomes large ($N \rightarrow \infty$), eq. (A1a) simplifies to

$$W(\mathbf{z}_\bullet, \mathbf{z}) = K(\mathbf{z}) \left[\left(1 + \alpha(\mathbf{z}_\bullet)(1 - 2\delta) \right) \frac{F(\mathbf{z}_\bullet)}{F(\mathbf{z})} + B(\mathbf{z}_\bullet, \mathbf{z}) \right], \quad (\text{A1c})$$

where

$$K(\mathbf{z}) = \lim_{N \rightarrow \infty} F(\mathbf{z})k(\mathbf{z}_\bullet, \mathbf{z}) = \frac{1}{2[1 - \delta \alpha(\mathbf{z})]} \quad (\text{A1d})$$

is a strictly positive term that captures the effect of density regulation, and

$$B(\mathbf{z}_\bullet, \mathbf{z}) = \lim_{N \rightarrow \infty} \left(\sum_{i=1}^{N-1} \beta(\mathbf{z}_\bullet, \mathbf{z}) \right) \quad (\text{A1e})$$

denotes the mutant's reproductive success as a pollen donor. Equation (A1c) corresponds to eq. (3) in the main text, and is used for all subsequent analyses.

A.1.2 Gradual evolution in a monomorphic population

Under the assumption that mutants are rare and have small, unbiased phenotypic effects, the gradual evolutionary dynamics of the population can be inferred by studying invasion fitness $W(\mathbf{z}_\bullet, \mathbf{z})$, and can be decomposed into two phases.

A.1.2.1 Directional selection

At first, a population monomorphic for some phenotype $\mathbf{z}$ will experience directional selection. The direction and strength of selection on a given trait z_v , $v \in \{1, \dots, V\}$, during this phase is given by the linear selection gradient on this trait,

$$S_v(\mathbf{z}) = \left. \frac{\partial W(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{v\bullet}} \right|_\circ, \quad (\text{A2a})$$

where the $|_\circ$ notation indicates that the derivative is evaluated at $\mathbf{z}_\bullet = \mathbf{z}$. A positive gradient means that alleles encoding a higher value of z_v are favoured by natural selection, whereas a negative gradient indicates that alleles encoding a lower value are favoured. In our model, this is

$$S_v(\mathbf{z}) = K(\mathbf{z}) \left[\frac{1}{F(\mathbf{z})} \left. \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{v\bullet}} \right|_\circ \left(1 + \alpha(\mathbf{z})(1 - 2\delta) \right) + (1 - 2\delta) \left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{v\bullet}} \right|_\circ + \left. \frac{\partial B(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{v\bullet}} \right|_\circ \right]. \quad (\text{A2b})$$

Equation (A2b) reveals that directional selection on a trait $z_v \in \mathbf{z}$ acts through three distinct pathways. The first pathway corresponds to the fitness effect of a change in z_v through its effect on ovule production. The term $\frac{1}{F(\mathbf{z})} \left. \frac{\partial F(\mathbf{z}_{\bullet})}{\partial z_v} \right|_0$ gives the effect of a small change in z_v on the relative production of ovules, meanwhile the term between brackets $1 + \alpha(\mathbf{z})(1 - 2\delta)$ gives the average number of mutant copies transmitted per ovule by the mutant for a given selfing rate $\alpha(\mathbf{z})$. The product of these two terms therefore gives the relative change in the number of mutant copies transmitted by the mutant through its own ovules, for a selfing rate $\alpha(\mathbf{z})$. The second pathway is the fitness effect of a change in z_v through its effect on the selfing rate. When inbreeding depression is low ($\delta < 1/2$), self-fertilised seeds transmit on average more mutant copies than the outcrossed and so selection will favour trait values that increase the selfing rate through this component. For example, if an increase in trait z_v results in a decrease in selfing rate ($\left. \frac{\partial \alpha(\mathbf{z}_{\bullet})}{\partial z_v} \right|_0 < 0$) then this effect favours lower values of z_v (the second component is negative). Conversely, when inbreeding depression is high ($\delta > 1/2$), outcrossed seeds transmit more mutant copies on average, so that selection favours trait values that decrease the selfing rate. The third and last pathway is the fitness effect of a change in z_v through its effect on its success as a pollen donor.

Through the sequential fixation of mutants of small effect, the population will travel in phenotypic space from one monomorphic state to the next in the direction favoured by selection, which is given by the vector of selection gradients $\mathbf{S}(\mathbf{z}) = (S_v(\mathbf{z}))_{1 \leq v \leq V}$. Eventually, it may reach a singular phenotype $\mathbf{z}^*$ for which directional selection vanishes[†], that is,

$$\mathbf{z}^* \quad \text{such that} \quad \mathbf{S}(\mathbf{z}) = \mathbf{0}. \quad (\text{A3})$$

Formally, whether the population will evolve towards a singular phenotype $\mathbf{z}^*$ can be assessed by computing the Jacobian matrix at this point, $\mathbf{J}(\mathbf{z}^*)$, which is a $V \times V$ matrix the (u, v) -entry $J_{uv}(\mathbf{z}^*)$ of which is given by

$$J_{uv}(\mathbf{z}^*) = \left. \frac{\partial S_u(\mathbf{z})}{\partial z_v} \right|_*, \quad (\text{A4})$$

where the notation $|_*$ indicates that the derivative is evaluated at the singular point (i.e., $\mathbf{z}_{\bullet} = \mathbf{z} = \mathbf{z}^*$).

[†]We note that in some cases, directional selection may also never vanish and drive a trait against a boundary of its definition domain. For example, for a trait $z_v \in [0, 1]$, selection may always favour a decrease in z_v such that $S_v(\mathbf{z}) < 0$ even once $z_v = 0$. In this case, $z_v = 0$ is stable so long as the population remains monomorphic for all other traits. If other traits undergo evolutionary branching irrespective of z_v (see below), the subsequent evolution of this trait requires additional analysis.

Convergence to the singular point $\mathbf{z}^*$ requires that the leading eigenvalue of $\mathbf{J}(\mathbf{z}^*)$ is negative under the assumption that all traits mutate independently (Leimar, 2009). In some cases, however, inspecting the selection gradients directly is sufficient to assess convergence stability.

A.1.2.2 Numerical analysis of evolution under directional selection

When several traits evolve jointly, explicit mathematical investigations of evolutionary dynamics under directional selection can become difficult. Nevertheless, useful insight can be gained from a numerical analysis where one follows the deterministic trajectory of a monomorphic population in phenotypic space, as dictated by selection gradients. Specifically, given a population monomorphic for phenotype $\mathbf{z}(\tau)$ at some time-step τ , we can compute the vector of selection gradients acting on the traits at that point in phenotypic space, $\mathbf{S}(\mathbf{z}(\tau))$, and use it to obtain the phenotype expressed by the population in the next time-step as

$$z_v(\tau + 1) = \mathcal{C}_{[z_v^{\min}, z_v^{\max}]} \left(z_v(\tau) + \Delta S_v(\mathbf{z}(\tau)) \right) \quad (\text{A5})$$

where $v \in \{1, \dots, V\}$, $0 < \Delta \ll 1$ is a small deviation and $\mathcal{C}_{[z_v^{\min}, z_v^{\max}]} : \mathbb{R} \rightarrow [z_v^{\min}, z_v^{\max}]$ denotes the clipping function that maintains trait z_v within its definition domain:

$$\mathcal{C}_{[z_v^{\min}, z_v^{\max}]}(z_v) = \begin{cases} z_v^{\min} & \text{if } z_v \leq z_v^{\min} \\ z_v & \text{if } z_v^{\min} < z_v < z_v^{\max} \\ z_v^{\max} & \text{if } z_v^{\max} \leq z_v. \end{cases} \quad (\text{A6})$$

Throughout, we assume that the small deviation Δ is a random variable

$$\Delta \sim \mathcal{U}(\Delta_{\min}, \Delta_{\max}), \quad (\text{A7})$$

where $\mathcal{U}(x, y)$ denotes the uniform distribution on $[x, y] \subset \mathbb{R}$ and $0 < \Delta_{\min} < \Delta_{\max}$ give the minimal and maximal possible deviations. This ensures that the algorithm does not remain trapped in stable oscillations around its equilibrium.

To determine the equilibrium of directional selection reached by a population initially monomorphic for some phenotype $\mathbf{z}(0)$, we can iterate eq. (A5) for all traits numerically until we reach a phenotype

$$\mathbf{z}^* \quad \text{such that} \quad |z_v(\tau + 1) - z_v(\tau)| < \epsilon \quad \text{for all} \quad z_v \in \mathbf{z}, \quad (\text{A8})$$

where $\epsilon > 0$ is a small tolerance criterion.

This algorithm is implemented in the *Mathematica* file named
`Numerical_DirectionalSelection.nb`, available at [10.5281/zenodo.23062245](https://zenodo.org/record/23062245).

A.1.2.3 Disruptive selection

Once the population expresses $\mathbf{z}^*$, leading order forces of selection vanish, so that second-order effects come into play. The nature of selection on the traits is then given by the Hessian matrix $\mathbf{H}(\mathbf{z}^*)$, which is a $V \times V$ matrix the (u, v) -entry $H_{uv}(\mathbf{z}^*)$ of which is given by

$$H_{uv}(\mathbf{z}^*) = \left. \frac{\partial^2 W(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{u\bullet} \partial z_{v\bullet}} \right|_{\mathbf{z}_\bullet = \mathbf{z} = \mathbf{z}^*}. \quad (\text{A9a})$$

Elements on the diagonal of $\mathbf{H}(\mathbf{z}^*)$ are given by the second derivative of $W(\mathbf{z}_\bullet, \mathbf{z})$ with respect to mutant trait value $H_{vv}(\mathbf{z}^*)$, evaluated at $\mathbf{z}_\bullet = \mathbf{z} = \mathbf{z}^*$, and therefore give the quadratic selection coefficient on z_v at the singular point. This coefficient captures the nature of selection acting on each trait independently. A negative value of $H_{vv}(\mathbf{z}^*)$ indicates that selection on trait z_v is stabilising and so favours a decrease in variance around z_v^* . A positive value, meanwhile, indicates that selection on z_v is disruptive at the singular point, and so favours an increase in variance from z_v^* . Off-diagonal elements (so where $u \neq v$) are the cross-derivative of $W(\mathbf{z}_\bullet, \mathbf{z})$ with respect to different trait values of the mutant. These elements correspond to the correlational selection coefficient between these two traits. A positive value of $H_{uv}(\mathbf{z}^*)$ indicates that selection favours a positive correlation between the two traits, whereas a negative $H_{uv}(\mathbf{z}^*)$ shows that selection favours a negative correlation. The evolution of a population expressing $\mathbf{z}^*$ is determined by the leading eigenvalue of $\mathbf{H}(\mathbf{z}^*)$. If it is negative, then selection is stabilising in all

dimensions of phenotype, favouring the population to remain monomorphic for $\mathbf{z}^*$, so that phenotypic evolution comes to a halt. If the leading eigenvalue is positive, on the other hand, disruptive selection will lead to the gradual emergence of polymorphism via evolutionary branching. It is useful to note here that there is two main ways for the leading eigenvalue of $\mathbf{H}(\mathbf{z}^*)$ to be positive. Either one diagonal element $H_{vv}(\mathbf{z}^*)$ is positive, meaning that one trait will undergo branching irrespective of selection on other traits, which may in turn cause other traits to become polymorphic; or all diagonal elements $H_{vv}(\mathbf{z}^*)$ are negative, so that selection would be stabilising for $\mathbf{z}^*$ if all traits evolved in isolation, but correlational selection amongst traits is sufficiently strong to overcome stabilising selection on each trait and lead to branching (D  barre et al., 2014). Thus, the more traits co-evolve, the more routes to polymorphism there is.

In the context of our model, the (u, v) -element of $\mathbf{H}(\mathbf{z}^*)$ associated with traits z_u and z_v , $H_{uv}(\mathbf{z}^*)$, is given by

$$\begin{aligned}
H_{uv}(\mathbf{z}^*) = K(\mathbf{z}) \times & \left[\frac{1}{F(\mathbf{z})} \frac{\partial^2 F(\mathbf{z}_\bullet)}{\partial z_{u\bullet} \partial z_{v\bullet}} \Big|_* \left(1 + \alpha(\mathbf{z})(1 - 2\delta) \right) \right. \\
& + (1 - 2\delta) \left(\frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{u\bullet} \partial z_{v\bullet}} \Big|_* + \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{u\bullet}} \Big|_* \times \frac{1}{F(\mathbf{z})} \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{v\bullet}} \Big|_* + \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{v\bullet}} \Big|_* \times \frac{1}{F(\mathbf{z})} \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{u\bullet}} \Big|_* \right) \\
& \left. + \frac{\partial^2 B(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{u\bullet} \partial z_{v\bullet}} \Big|_* \right].
\end{aligned}
\tag{A9b}$$

Equation (A9b) shows that second-order forces of selection act through similar pathways to directional selection. The first term between square brackets captures the fitness effect of a joint change in z_u and z_v via its effect on ovule production, for a given selfing rate. The second term captures the fitness effect of a joint change in z_u and z_v through the change it causes in the number of self-fertilised ovules, which can either result from the joint effect of z_u and z_v on the selfing rate, or from the effect of one trait on ovule production and of the other on the selfing rate. Similar to the directional selection case, the direction of this effect depends on whether outcrossed or selfed seeds yield higher fitness returns, i.e., on whether inbreeding depression is greater or lower than a half, respectively. The last term captures the effect of a joint change in z_u and z_v on reproductive success as a pollen donor.

A.1.3 The nature of the polymorphism resulting from evolutionary branching

A.1.3.1 Analytical setup

When polymorphism emerges through evolutionary branching, the nature of the polymorphism maintained at equilibrium can be studied by modelling a dimorphic resident population in which two types of individuals, say A and B expressing strategies $\mathbf{z}_A$ and $\mathbf{z}_B$, respectively, coexist at an ecological equilibrium. Under the assumption that the population is large ($N \rightarrow \infty$), this equilibrium can be described by the equilibrium frequency of type A,

$$\hat{p}(\mathbf{z}_A, \mathbf{z}_B) \quad \text{such that} \quad \Delta p(\mathbf{z}_A, \mathbf{z}_B) = p_{t+1} - p_t = 0, \quad (\text{A10a})$$

where p_t is the frequency of type A at time t and $\Delta p(\mathbf{z}_A, \mathbf{z}_B)$ is the change in this frequency over one generation. In our model, the frequency change is given by

$$\begin{aligned} \Delta p(\mathbf{z}_A, \mathbf{z}_B) = & \frac{2F(\mathbf{z}_A)p_t}{\bar{F}(\mathbf{z}_A, \mathbf{z}_B)} \left[\alpha(\mathbf{z}_A)(1 - \delta) + \beta(\mathbf{z}_A, \mathbf{z}_A)p_t + \frac{1}{2}\beta(\mathbf{z}_B, \mathbf{z}_A)(1 - p_t) \right] \\ & + \frac{2F(\mathbf{z}_B)(1 - p_t)}{\bar{F}(\mathbf{z}_A, \mathbf{z}_B)} \frac{1}{2}\beta(\mathbf{z}_A, \mathbf{z}_B)p_t - p_t, \end{aligned} \quad (\text{A10b})$$

where

$$\bar{F}(\mathbf{z}_A, \mathbf{z}_B) = 2F(\mathbf{z}_A)[1 - \delta\alpha(\mathbf{z}_A)]p_t + 2F(\mathbf{z}_B)[1 - \delta\alpha(\mathbf{z}_B)](1 - p_t) \quad (\text{A10c})$$

is the average number of mature seeds produced per individual in the population at time t .

The term on the first line in eq. (A10b) represents the proportion of the next generation that carries $\mathbf{z}_A$ and has a type A individual as maternal parent. The ratio on the left-hand side gives proportion of seeds in the next generation produced by these parents, and the term between square brackets gives the probability that these seeds carry $\mathbf{z}_A$ depending on how they were fertilised. Each type A individual self-fertilises at rate $\alpha(\mathbf{z}_A)$, in which case the resulting diploid zygote survives with probability $1 - \delta$ and divides to produce two seeds carrying $\mathbf{z}_A$; a fraction $\beta(\mathbf{z}_A, \mathbf{z}_A)p_t$ of its ovules is fertilised by outcross pollen carrying $\mathbf{z}_A$, in which case the diploid zygote survives with probability one and yields only seeds carrying $\mathbf{z}_A$; finally,

a fraction $\beta(\mathbf{z}_B, \mathbf{z}_A)(1 - p_t)$ of its ovules are fertilised by $\mathbf{z}_B$ pollen and the resulting zygote survives with probability one, dividing to yield one seed carrying $\mathbf{z}_A$ and the other carrying $\mathbf{z}_B$. The term on the second line gives the proportion of the population in the next generation that carries $\mathbf{z}_A$ and has a type B individual as maternal parent. Similar to type A maternal parents, the ratio on the left-hand side gives the proportion of seeds in the next generation produced by these parents. It is multiplied by the probability that these seeds carry $\mathbf{z}_A$, which is only the case for 1/2 of the seeds produced through outcrossing with $\mathbf{z}_A$ -bearing pollen, which occurs in proportion $\beta(\mathbf{z}_A, \mathbf{z}_B)p_t$.

In a polymorphic population at its ecological equilibrium (as given by eq. A10), we now consider the fate of a rare mutant $\mathbf{z}_\bullet$ arising as a single copy, which is again determined by its invasion fitness, denoted as $W_p(\mathbf{z}_\bullet, \mathbf{z}_A, \mathbf{z}_B)$ to signify that the mutant $\mathbf{z}_\bullet$ invades into a population that is polymorphic for $\mathbf{z}_A$ and $\mathbf{z}_B$. Under the assumption that the mutant deviates weakly of either type A or type B residents, the mutant's invasion fitness can be used to obtain the selection gradient on each resident's phenotype. Specifically, the selection gradient on the value of trait z_v encoded by resident type $x \in \{A, B\}$, $z_{v,x}$ can be obtained as

$$S_{v,x}(\mathbf{z}_A, \mathbf{z}_B) = \left. \frac{\partial W_p(\mathbf{z}_\bullet, \mathbf{z}_A, \mathbf{z}_B)}{\partial z_{v,\bullet}} \right|_{\mathbf{z}_\bullet = \mathbf{z}_x}. \quad (\text{A11})$$

Similar to the monomorphic case, a positive selection gradient indicates that selection favours an increase in trait $z_{v,x}$ from its current value, whereas a negative gradient indicates that a decrease in trait $z_{v,x}$ is favoured by selection. In this setting, small-effect mutants arising either from the type A or the type B allele will be able to invade and replace their ancestral allele if they encode a phenotype taking their type further in the direction favoured by selection, given by the type-specific vectors of selection gradients $\mathbf{S}_A(\mathbf{z}_A, \mathbf{z}_B) = (S_{v,A}(\mathbf{z}_A, \mathbf{z}_B))_{1 \leq v \leq V}$ and $\mathbf{S}_B(\mathbf{z}_A, \mathbf{z}_B) = (S_{v,B}(\mathbf{z}_A, \mathbf{z}_B))_{1 \leq v \leq V}$.

Eventually, this invasion-and-replacement process may lead the population to an equilibrium coalition where the gradients vanish, i.e.,

$$\mathbf{z}_A^*, \mathbf{z}_B^* \quad \text{such that} \quad \mathbf{S}_A(\mathbf{z}_A^*, \mathbf{z}_B^*) = \mathbf{S}_B(\mathbf{z}_A^*, \mathbf{z}_B^*) = 0, \quad (\text{A12})$$

or push traits against a boundary of their definition domain (e.g., $S_{v,x}(\mathbf{z}_A, \mathbf{z}_B) < 0$ when $z_{v,x} = 0$),

so that evolution owing to directional selection on each type comes to a halt. Determining whether this equilibrium coalition is evolutionarily stable (in the sense that neither of the two branches undergo another round of branching once they sit at an equilibrium of directional selection $\mathbf{z}_x^*$) would require computing type-specific Hessian matrices, but we leave this aside here.

A.1.3.2 Numerical analyses

It is usually difficult to solve explicitly for the equilibrium polymorphic coalition favoured by natural selection, especially if several traits evolve jointly. We therefore resort to a numerical analysis in which we follow the deterministic trajectories of the phenotypes expressed by type A and B from an arbitrary initial coalition $(\mathbf{z}_A, \mathbf{z}_B)$ across phenotypic space, as dictated by the vectors of gradients $\mathbf{S}_A(\mathbf{z}_A, \mathbf{z}_B)$ and $\mathbf{S}_B(\mathbf{z}_A, \mathbf{z}_B)$. Specifically, given the phenotypes expressed by type A and B in the population at some time-step τ , $\mathbf{z}_A(\tau)$ and $\mathbf{z}_B(\tau)$, the elements $z_{v,x}(\tau + 1)$ of the phenotype $\mathbf{z}_x(\tau + 1)$ expressed by type x in the next time-step are obtained as

$$z_{v,x}(\tau + 1) = \mathcal{C}_{[z_v^{\min}, z_v^{\max}]} \left(z_{v,x}(\tau) + \Delta S_{v,x}(\mathbf{z}_A(\tau), \mathbf{z}_B(\tau)) \right) \quad (\text{A13})$$

where $0 < \Delta \ll 1$ is a small deviation and $\mathcal{C}_{[z_v^{\min}, z_v^{\max}]} : \mathbb{R} \rightarrow [z_v^{\min}, z_v^{\max}]$ denotes the clipping function, similar to the monomorphic case described in section A.1.2.2. To avoid oscillatory behaviours, we also assume here that the small deviation Δ is a random variable

$$\Delta \sim \mathcal{U}(\Delta_{\min}, \Delta_{\max}). \quad (\text{A14})$$

To determine the evolutionary equilibrium reached by a dimorphic population, we can iterate eq. (A13) for all traits numerically until we reach

$$\mathbf{z}_A^*, \mathbf{z}_B^* \quad \text{such that} \quad |z_{v,A}(\tau + 1) - z_{v,A}(\tau)| < \epsilon \quad \text{and} \quad |z_{v,B}(\tau + 1) - z_{v,B}(\tau)| < \epsilon \quad (\text{A15})$$

for all $z_{v,A} \in \mathbf{z}_A$ and $z_{v,B} \in \mathbf{z}_B$, where $\epsilon > 0$ is a small tolerance criterion, similar to the monomorphic case (Section A.1.2.2); or until one type displaces the other (i.e., until the equilibrium frequency in the

population defined by eq. (A10a) hits $\hat{p}(\mathbf{z}_A, \mathbf{z}_B) = 0$ or 1), at which point polymorphism is lost so the population is again monomorphic and evolving under directional selection.

There is a couple of options for how to initialise the numerical recursion described above. A first option is to compute the leading eigenvector of the Hessian matrix $\mathbf{H}(\mathbf{z}^*)$ (eq. A9a) at the singular point, $\boldsymbol{\eta}(\mathbf{z}^*)$. This vector gives the direction in which branching initially takes place in phenotypic space. We may then position the two types along this vector in the close vicinity of the singular point $\mathbf{z}^*$, i.e.,

$$\mathbf{z}_A(\tau = 0) = \mathbf{z}^* + \epsilon \boldsymbol{\eta}(\mathbf{z}^*) \quad \text{and} \quad \mathbf{z}_B(\tau = 0) = \mathbf{z}^* - \epsilon \boldsymbol{\eta}(\mathbf{z}^*), \quad (\text{A16})$$

where $0 < \epsilon \ll 1$ is a small constant. Iterating eq. (A13) for all traits from this starting point gives the deterministic trajectory that evolutionary branching is expected to follow from this singular point.

This first option works for most of the parameter space. However, for some parameter values for which selection is fairly weak, the two types may take a very long time to diverge and reach their respective equilibrium. To circumvent this issue and save computation time, a second option is to start the algorithm with strongly diverged types in the direction given by the eigenvector $\boldsymbol{\eta}(\mathbf{z}^*)$, provided that such polymorphic situation is possible (both types are maintained in a stable coalition). The algorithm then rapidly converges to an equilibrium, because greater phenotypic differences between types tend amplify selection when the branches have not yet reached their respective equilibria. We ran the algorithm using both initialisations whenever possible for our models, and found that the same equilibrium was always reached.

This algorithm is implemented in the *Mathematica* file named
 Numerical_DisruptiveSelection.nb, available at [10.5281/zenodo.23062245](https://zenodo.org/record/23062245).

A.2 The joint evolution of sex allocation and pollen export

In this section, we derive the analytical results presented in Section 3 of the main text. We consider the joint evolution of two traits, sex allocation $z_1 \in [0, 1]$, which determines the proportion of its reproductive resources that an individual commits to its female function, and pollen export $z_2 \in [0, 1]$, which determines the proportion of its pollen that an individual disperses to fertilise ovules on other plants. The population follows the life cycle described in Section A.1.1, and the two traits affect the behaviour of individuals along this life cycle as follows.

(i) *Sexual development*: The number of ovules and pollen grains produced by individual i , $F(\mathbf{z}_i)$ and $M(\mathbf{z}_i)$ are given by

$$F(\mathbf{z}_i) = F_0 z_{1,i} \quad \text{and} \quad M(\mathbf{z}_i) = M_0 (1 - z_{1,i}), \quad (\text{A17})$$

respectively, where $F_0 \gg 0$ is the maximum female fecundity (the number of ovules produced by an individual allocating all its resources to its female function) and $M_0 \gg F_0$ is the maximum male fecundity. Thus, ovule and pollen production depend linearly on resource allocation, and pollen export does not influence gamete production.

(ii) *Mating*: After sexual development, plant i disperses a fraction $z_{2,i}$ of its pollen to fertilise ovules on other plants, and retains the remainder fraction $1 - z_{2,i}$ for self-fertilisation, following a mass-action model (Holsinger, 1991; Porcher and Lande, 2005). A fraction $c \in [0, 1]$ of the dispersed pollen is lost during dispersal, and the successful fraction $1 - c$ is distributed equally amongst the $N - 1$ other plants. As a result, the selfing rate $\alpha(\mathbf{z}_i)$ of individual i is given by the proportion of its own pollen out of the total amount of pollen it received, i.e.,

$$\alpha(\mathbf{z}_i) = \frac{M(\mathbf{z}_i) (1 - z_{2,i})}{M(\mathbf{z}_i) (1 - z_{2,i}) + \frac{1 - c}{N - 1} \sum_{j \neq i} M(\mathbf{z}_j) z_{2,j}}, \quad (\text{A18a})$$

and the probability that individual i sires an ovule of individual $j \neq i$ is given by the share of pollen from

i out of all the pollen received by j ,

$$\beta(\mathbf{z}_i, \mathbf{z}_j) = \frac{z_{2,i} \frac{1-c}{N-1} M(\mathbf{z}_i)}{M(\mathbf{z}_j)(1-z_{2,j}) + \frac{1-c}{N-1} \sum_{k \neq j} z_{2,k} M(\mathbf{z}_k)}. \quad (\text{A18b})$$

The rest of the life cycle proceeds as explained in Section A.1.1. Self-fertilised zygotes suffer from inbreeding depression, becoming viable with probability $1 - \delta$, and viable diploid zygotes immediately undergo meiosis to give rise to two haploid seeds each. Then, all adults die and the next generation is sampled from the produced seeds.

A.2.1 Directional selection

Using eq. (A2), the selection gradient on pollen export z_2 , is given by

$$S_2(\mathbf{z}) = K(\mathbf{z}) \left(\left. \frac{\partial B(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{2\bullet}} \right|_o + (1-2\delta) \left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \right|_o \right) = \underbrace{K(\mathbf{z}) \frac{1-c}{(1-cz_2)^2}}_{>0} \left(1 - z_2(1+c-2\delta) \right), \quad (\text{A19a})$$

which is independent of sex allocation, as pointed out by Zhang Zhang (2000). This is because the selection gradient captures directional selection on each trait holding everything else fixed, in which case all individuals have the same pollen production $M(\mathbf{z})$, which simplifies away in eq. (A18).

Pollen export z_2 does not affect ovule and pollen production ($\left. \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \right|_o = \left. \frac{\partial M(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \right|_o = 0$). The selection gradient in eq. (A19a) therefore reflects a balance between the fitness consequences of a change in pollen export through its effect on siring success as a pollen donor and the selfing rate, with the direction of selection on pollen export dictated by the term in brackets on the right-hand side of eq. (A19a). This term is linear in z_2 , and upon closer inspection reveals that one of two things can happen depending on the cost of dispersal c and inbreeding depression δ . Either inbreeding depression is strong relative to the cost of dispersal, specifically,

$$\delta \geq \frac{c}{2}, \quad (\text{A19b})$$

in which case eq. (A19a) is positive for all $z_2 \in [0, 1]$, so that selection favours complete pollen dispersal

($z_2 \rightarrow 1$), leading the population to become completely outcrossing; or inbreeding depression is weak relative to the costs of dispersal ($\delta < c/2$), in which case there exists an intermediate pollen export rate

$$z_2^* = \frac{1}{1 + c - 2\delta} \quad (\text{A19c})$$

at which directional selection vanishes. The term between brackets is then strictly positive at $z_2 = 0$ and negative at $z_2 = 1$. Given it is linear in z_2 , this implies that the singular pollen export rate in eq. (A19c) is always convergence stable. The singular export rate is a decreasing function of the cost of dispersal: the more pollen gets lost during export, the lower selection for pollen dispersal because its benefits in terms of siring success as pollen donor are reduced. Eq. (A19c) is also an increasing function of inbreeding depression: the higher inbreeding depression, the lower the benefits of selfing are, which promotes higher dispersal.

The selection gradient on sex allocation z_1 , meanwhile, is given by

$$S_1(\mathbf{z}) = K(\mathbf{z}) \left[\underbrace{\left(1 + (1 - 2\delta)\alpha(\mathbf{z})\right) \frac{1}{z_1}}_{\text{Effect on seed production}} - \underbrace{(1 - 2\delta)\alpha(\mathbf{z}) (1 - \alpha(\mathbf{z})) \frac{1}{1 - z_1}}_{\text{Effect on selfing rate}} - \underbrace{(1 - \alpha(\mathbf{z})) \frac{1}{1 - z_1}}_{\text{Effect on success as pollen donor}} \right] \quad (\text{A20a})$$

where

$$\alpha(\mathbf{z}) = \frac{1 - z_2}{1 - cz_2} \quad (\text{A20b})$$

is the selfing rate in the resident population, which does not depend on sex allocation because all individuals produce the same amount of pollen so that $M(\mathbf{z})$ cancels out in eq. (A18a).

Equation (A20a) shows that directional selection on sex allocation is shaped through the three pathways described in Section A.1.2, which all depend on the selfing rate $\alpha(\mathbf{z})$. The first term, which gives the fitness consequences of a change in sex allocation through its effect on seed production, is always positive (increasing allocation to female function always increases fitness returns through this function). Its influence on sex allocation evolution is amplified by selfing when inbreeding depression is weak ($\delta < 1/2$): it selects more strongly for increased allocation to female function, because selfing increases the fitness

returns per seed in this case. Conversely, selfing tends to reduce selection for allocation to female function through this effect under strong inbreeding depression. The second term gives the fitness consequences of a change in sex allocation through its effect on the selfing rate, which depend on the magnitude of inbreeding depression as well. This term is negative when selfing is beneficial ($\delta < 1/2$): it promotes greater allocation to male function because this increases the selfing rate; and positive when selfing is detrimental ($\delta > 1/2$), promoting greater allocation to female function because this reduces the selfing rate. Finally, the third term gives the fitness effect of a change in sex allocation through its effect on success as a pollen donor. It is always negative, therefore promoting allocation to male function (producing more pollen always has a positive impact on fitness returns as a pollen donor), but is always reduced by selfing. This is because the more individuals retain pollen for selfing, the less likely it is that an outcrossing pollen donor will be able to fertilise them.

Re-arranging eq. (A20a), we obtain

$$S_1(\mathbf{z}) = K(\mathbf{z}) \left(1 + (1 - 2\delta)\alpha(\mathbf{z}) \right) \left[\frac{1}{z_1} - (1 - \alpha(\mathbf{z})) \frac{1}{1 - z_1} \right], \quad (\text{A20c})$$

which tells us that the opposing effects of selfing on fitness returns through a focal individual's own ovules (the first two terms in eq. A20a) exactly balance out: any increase in ovule production is met with a proportional decrease in the expected fitness return per ovule. Thus, only the effect of selfing on success as a pollen donor remains, which tends to favour female-biased sex allocations. Solving for the singular sex allocation (eq. (A3), we find

$$S_1(\mathbf{z}^*) = 0 \quad \Leftrightarrow \quad z_1^* = \frac{1}{2 - \alpha(\mathbf{z}^*)}, \quad (\text{A20d})$$

in agreement with previous analyses (e.g., eq. 31 with $n = 1$ in Zhang (2000); also derived in Lesaffre et al. (2025)).

Combining this result with our findings on pollen export rate (eq. A19), we have that directional selection either favours complete outcrossing ($z_2 \rightarrow 1$ when $\delta \geq c/2$), in which case it also favours equal investment to male and female function ($z_1^* = 1/2$); or it favours an intermediate export rate $z_2^* = 1/(1 + c - 2\delta)$,

resulting in mixed mating and a female-biased sex allocation.

A.2.2 Disruptive selection

When inbreeding depression is sufficiently strong for selection to favour complete outcrossing, pollen export is under persistent directional selection to remain complete ($S_2(\mathbf{z}) > 0$ at $z_2 = 1$), so that the model collapses to a randomly mating hermaphroditic population with linear fitness returns through each sexual function. In this situation, equal investment into male and female function, $z_1^* = 1/2$, constitutes an evolutionary stable strategy (Charnov, 1982; Lesaffre et al., 2024, 2025). We therefore do not consider this case further.

In contrast, when inbreeding depression is such that an intermediate rate of pollen export is favoured, directional selection vanishes on both traits at the singular point $\mathbf{z}^* = (z_1^*, z_2^*)$ so that second-order forces as described by the Hessian matrix at $\mathbf{z}^*$ dictate evolutionary dynamics (Section A.1.2.3 above).

Using eq. (A9b), the quadratic selection coefficient on sex allocation, $H_{11}(\mathbf{z}^*)$, is given by

$$H_{11}(\mathbf{z}^*) = \underbrace{K(\mathbf{z}^*)(1 - 2\delta)}_{\text{Positive when } \delta < c/2} \left[\left. \frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet}^2} \right|_* + 2 \left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet}} \right|_* \times \frac{1}{F(\mathbf{z}^*)} \left. \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{1\bullet}} \right|_* \right]. \quad (\text{A21a})$$

The term between square brackets depends on two elements. The first is the second derivative of the selfing rate with respect to mutant sex allocation. This is strictly negative, because the individual's own pollen increasingly competes with itself for self-fertilisation as the amount produced increases, leading to diminishing returns on the selfing rate (analogous to a form of local sperm/mate competition, Hamilton, 1967; Schärer, 2009). The second term captures the joint effect of a change in sex allocation on ovule production and the selfing rate. Since any change in sex allocation has opposite effects on these two quantities, this term is necessarily negative as well. Thus, eq. (A21a) as a whole is negative, indicating that partial selfing tends to generate stabilising selection on sex allocation under weak inbreeding depression (Lesaffre et al., 2025).

The quadratic selection coefficient on pollen export, $H_{22}(\mathbf{z}^*)$, is given by

$$H_{22}(\mathbf{z}^*) = \underbrace{K(\mathbf{z}^*)(1 - 2\delta)}_{>0} \frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}^2} \Big|_*, \quad (\text{A21b})$$

so that its sign is determined by the sign of $\frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}^2} \Big|_*$, which is also negative for reasons similar to those exposed above: as the proportion of pollen exported decreases, competition amongst the individual's own pollen grains intensifies and leads to diminishing returns in terms of increased selfing. Quadratic selection on pollen export is therefore also stabilising.

Finally, the correlational selection coefficient between sex allocation and pollen export, $H_{12}(\mathbf{z}^*)$, is

$$H_{12}(\mathbf{z}^*) = \underbrace{K(\mathbf{z}^*)}_{>0} \left[\frac{\partial^2 B(\mathbf{z}_\bullet, \mathbf{z})}{\partial z_{1\bullet} \partial z_{2\bullet}} \Big|_* + (1 - 2\delta) \left(\frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet} \partial z_{2\bullet}} \Big|_* + \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \Big|_* \times \frac{1}{F(\mathbf{z}^*)} \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{1\bullet}} \Big|_* \right) \right]. \quad (\text{A21c})$$

The first term in square brackets captures the effect of the interaction between the two traits on success as a pollen donor, which is always negative. This is because increasing allocation to female function z_1 always has a negative impact on success as a pollen donor, and this impact becomes increasingly strong (i.e., more negative) as the focal individual's pollen export increases. In other words, a higher pollen export rate makes a decrease in pollen production more negative for siring success as a pollen donor. This term therefore tends to favour a negative correlation between sex allocation and pollen export.

The second term is proportional to $1 - 2\delta$, capturing the effect of the interaction between z_1 and z_2 on fitness gains through the focal individual's own ovules, which is decomposed in two parts. The first part is the cross-derivative of the selfing rate with respect to both traits $\frac{\partial^2 \alpha(\mathbf{z}_\bullet)}{\partial z_{1\bullet} \partial z_{2\bullet}} \Big|_*$, which captures the effect of the interaction between the two traits on the selfing rate of a focal individual. This term is positive when the singular pollen export rate is above a threshold, specifically such that

$$\alpha(\mathbf{z}^*) < \frac{1}{2} \quad \Leftrightarrow \quad c < \frac{1 + 2\delta}{2}, \quad (\text{A21d})$$

which promotes a positive association between sex allocation and pollen export. This is because in this case, the pollen retained by the individual makes up the minority of the total amount of pollen that it

receives, so that increasing pollen export has a small impact on this total amount. Increasing the pollen export rate then tends to decrease the negative impact that increasing the focal individual's allocation to female function has on its own selfing rate, which favours a positive correlation between the two traits (the contribution of the individual's own pollen to the denominator in eq. A18a is relatively small, so that the effect of traits changes is mostly driven by the numerator). Conversely, this term promotes a negative association between sex allocation and pollen export when the pollen export rate is below the threshold. This is because an individual's own pollen then makes up the majority of the total amount of pollen it receives, so that increasing its pollen export rate decreases both the amount of pollen it retains and the total amount of pollen received, which overall tends to amplify the effect of the focal's sex allocation on its selfing rate. The second part captures the joint effect of a change in pollen export on the selfing rate ($\left. \frac{\partial \alpha(\mathbf{z}_\bullet)}{\partial z_{2\bullet}} \right|_0$) and a change in sex allocation on relative ovule production ($\left. \frac{\partial F(\mathbf{z}_\bullet)}{\partial z_{1\bullet}} \right|_0 / F(\mathbf{z}^*)$) because increasing pollen export has a negative effect on the selfing rate and increasing sex allocation has a positive effect on ovule production, this part is negative and therefore promotes a negative correlation between pollen export and sex allocation. Overall, plugging eq. (A18) into eq. (A21c), the correlation selection coefficient simplifies to

$$H_{12}(\mathbf{z}^*) = -\frac{(1-c)(1+c-2\delta)}{2z_1^*(1-z_1^*)[1-\delta(c+2(1-\delta))]} \left(1 + \frac{1-2(c-\delta)}{c-2(1-\delta)} \right), \quad (\text{A21e})$$

which is strictly negative for $\delta < c/2$, indicating that correlational selection on sex allocation and pollen export overall favours a negative correlation between the two traits, i.e., it favours individuals that allocate more to their male function to export pollen at a higher rate, and vice-versa.

Putting the elements of the Hessian matrix together (eq. A21), we compute the characteristic polynomial $\mathbf{H}(\mathbf{z}^*)$, the roots of which give the eigenvalues of this matrix and therefore inform us on the nature of selection at $\mathbf{z}^*$. We refrain from giving its full expression here as it is unsightly, but it is available in the accompanying *Mathematica* notebook (`AnalyticalResults.nb` available at [10.5281/zenodo.23062245](https://zenodo.org/record/23062245)). The polynomial takes the form

$$\mathcal{P}(\lambda) = \lambda^2 + \gamma_1 \lambda + \gamma_0, \quad (\text{A22})$$

and is therefore an upward-facing parabola, as its quadratic coefficient is positive. This implies that the parabola is negative between its two roots. Thus, if it is negative at $\lambda = 0$, then exactly one eigenvalue of $\mathbf{H}(\mathbf{z}^*)$ is positive. Evaluating $\mathcal{P}(\lambda)$ at this point we have

$$\mathcal{P}(0) = \gamma_0 = - \left[\frac{(1-c)(1+c-2\delta)(c-2(1-\delta))}{2(1-2\delta)[1-\delta(c+2(1-\delta))]} \right]^2 < 0. \quad (\text{A23})$$

The Hessian matrix $\mathbf{H}(\mathbf{z}^*)$ therefore always has a positive eigenvalue despite having a strictly negative diagonal when $\delta < c/2$ (Fig. 1A). Biologically, this implies that selection is always disruptive on sex allocation and pollen export as a result of correlational selection, favouring the emergence of two types of individuals: one that allocates more resources to its female function while reducing its pollen export rate, and one that allocates more to its male function with an increased pollen export (Fig. 1B). In other words, selection favours the emergence of one type that specialises in the production of ovules and self-fertilisation, and another that specialises into pollen production and export.

A.2.3 Polymorphism resulting from evolutionary branching

We ran numerical analyses as described in Section A.1.3.2 to investigate the nature of polymorphism maintained at evolutionary equilibrium. We find that evolutionary branching results in the coexistence of two types, a pure male type that exports all its pollen ($z_1^* = 0$ and $z_2^* = 1$) and a hermaphroditic type that exports none of its pollen ($z_2^* = 0$), leading to a form of androdioecy. See main text for more details.

A.3 Joint evolution with self-pollen advantage

In this section, we derive the analytical results presented in Section 4 of the main text. We investigate the joint evolution of three traits: sex allocation z_1 and the pollen export rate z_2 , as in the previous section, and the self-pollen advantage, $z_3 \in [0, 1]$ which controls the competitive advantage that a plant grants its own pollen relative to pollen from other donors. The population follows the life cycle described in Section A.1.1, and the three traits affect the behaviour of individuals along this life cycle as follows.

(i) *Sexual development*: The number of ovules and pollen grains produced by individual i are given by

$$F(\mathbf{z}_i) = F_0 z_{1,i} \quad \text{and} \quad M(\mathbf{z}_i) = M_0(1 - z_{1,i}), \quad (\text{A24a})$$

respectively. We assume that the number of ovules is large relative to the number of breeding spots ($N \ll F_0$), and that there is always enough pollen for all ovules to be fertilised ($F_0 \ll M_0$), i.e., no pollen limitation.

(ii) *Mating*: After sexual development, individuals export a fraction $z_2 \in [0, 1]$ of their pollen to fertilise ovules on other plants, and the rest $(1 - z_2)$ remains on the plant for self-fertilisation (Holsinger, 1991; Zhang, 2000; Porcher and Lande, 2005). A fraction $c \in [0, 1]$ of the exported pollen is lost during dispersal, and the successful fraction $1 - c$ is distributed equally among the $N - 1$ other plants. The self- and cross-pollen received by each plant then compete for ovule fertilisation, and this competition is biased by the maternal plant either in favour of or against its own pollen according to its self-pollen advantage $z_{3,i}$. Competition is unbiased when $z_{3,i} = 1/2$, so that pollen grains from all sources are equally competitive, whereas $z_{3,i} > 1/2$ and $z_{3,i} < 1/2$ favour and oppose self-fertilisation, respectively. The self-pollen advantage z_3 thus modulates the relative competitiveness of self- and cross-pollen grains for fertilisation. Overall, the selfing rate $\alpha(\mathbf{z}_i)$ of individual i is given by

$$\alpha(\mathbf{z}_i) = \frac{z_{3,i} M_{\text{self}}(\mathbf{z}_i)}{z_{3,i} M_{\text{self}}(\mathbf{z}_i) + (1 - z_{3,i}) \times \underbrace{\frac{1 - c}{N - 1} \sum_{j \neq i} M_{\text{out}}(\mathbf{z}_j)}_{\text{Total amount of cross-pollen received by plant } i}}, \quad (\text{A24b})$$

and the probability that individual i fertilises an ovule of individual $j \neq i$ is

$$\beta(\mathbf{z}_i, \mathbf{z}_j) = \frac{(1 - z_{3,j}) \frac{1 - c}{N - 1} M_{\text{out}}(\mathbf{z}_i)}{z_{3,j} M_{\text{self}}(\mathbf{z}_j) + (1 - z_{3,j}) \frac{1 - c}{N - 1} \sum_{k \neq j} M_{\text{out}}(\mathbf{z}_k)}, \quad (\text{A24c})$$

where

$$M_{\text{self}}(\mathbf{z}_i) = M(z_{1,i})(1 - z_{2,i})P_{\text{self}} \quad \text{and} \quad M_{\text{out}}(\mathbf{z}_i) = M(z_{1,i})z_{2,i}P_{\text{out}}(z_{3,i}) \quad (\text{A24d})$$

respectively give the amount of self-pollen transferred to plant i 's own stigmas and the amount of pollen exported away by plant i . In eq. (A24d), $P_{\text{self}} \in [0, 1]$ is a parameter that captures the efficiency of within-plant pollen transfers, while $P_{\text{out}}(z_{3,i}) \in [0, 1]$ is function that captures the efficiency of pollen export, which decreases with the individual's self-pollen advantage $z_{3,i}$, reflecting sexual interference and pollen discounting.

Since $M_{\text{self}}(\mathbf{z}_i)$ and $M_{\text{out}}(\mathbf{z}_i)$ appear exclusively in ratios in eqs. (A24b) and (A24c), only their relative scaling matters. Without loss of generality, we may therefore divide both by P_{self} so that they become

$$M_{\text{self}}(\mathbf{z}_i) = M(z_{1,i})(1 - z_{2,i}) \quad \text{and} \quad M_{\text{out}}(\mathbf{z}_i) = M(z_{1,i})z_{2,i}P(z_{3,i}), \quad (\text{A24e})$$

where $P(z_{3,i}) = P_{\text{out}}(z_{3,i})/P_{\text{self}}$ is the relative efficiency of pollen export, which we will work with from now on.

The rest of the life cycle proceeds as explained in Section A.1.1. Self-fertilised zygotes suffer from inbreeding depression, becoming viable with probability $1 - \delta$, and viable diploid zygotes immediately undergo meiosis to give rise to two haploid seeds each. Then, all adults die and the next generation is sampled from the produced seeds.

A.3.1 Cost-free self-pollen promotion ($P(z_3) = 1$)

We first consider the case where self-pollen promotion causes no sexual interference, so that $P(z_3) = 1$ for all $\mathbf{z}$. In this case, we find that the selection gradient on self-pollen advantage is given by

$$S_3(\mathbf{z}) = K(\mathbf{z}) \frac{z_2(1-c)(1-z_2)}{[z_2(1-c)(1-z_3) + (1-z_2)z_3]^2} (1-2\delta), \quad (\text{A25a})$$

which does not depend on sex allocation z_1 for reasons similar to the pollen export rate above (see eq. A19 for explanation). Eq. (A25a) shows that the self-pollen advantage is only under selection when the pollen export rate is intermediate (i.e., when $0 < z_2 < 1$). This is because self-pollen never competes with cross-pollen otherwise, so that z_3 is effectively neutral here. Provided that $0 < z_2 < 1$, eq. (A25a) also shows

that there can be no intermediate equilibrium for the self-pollen advantage under directional selection, as the gradient never changes sign for $z_3 \in [0, 1]$. Thus, at an equilibrium of directional selection, either $z_3^* = 0$ and the population is strictly outcrossing (self-pollen cannot fertilise) unless no pollen is exchanged ($z_2 = z_3 = 0$) which is a degenerate case where the population cannot persist, as no ovule gets fertilised; or $z_3^* = 1$ and the population is strictly selfing (only self-pollen can fertilise) unless individuals export all their pollen ($z_2 = z_3 = 1$) which is also a degenerate case where no ovule gets fertilised and the population goes extinct. Which of these two equilibria will be reached by an evolving population is determined by inbreeding depression. Specifically, either $\delta < 1/2$ and eq. (A25a) is positive, so that a population with an intermediate initial selfing rate will evolve towards complete selfing ($z_3 \rightarrow 1$); or $\delta > 1/2$ and eq. (A25a) is negative, so that the population will evolve complete outcrossing ($z_3 \rightarrow 0$). The $\delta = 1/2$ threshold comes from the two-fold transmission advantage of selfing which must be overcome by inbreeding depression for outcrossing to be favoured, as shown by classical theory in the absence of sexual interference (Lande and Schemske, 1985).

Once the population has become either completely outcrossing or completely selfing, the joint evolution of sex allocation and pollen export is trivial. Under enforced outcrossing ($z_3 = 0$), the selection gradient on pollen export reduces to

$$S_2(\mathbf{z}) = \frac{K(\mathbf{z})}{z_2}, \quad (\text{A25b})$$

which is always positive, and so favours complete pollen dispersal. This is because any pollen that is not dispersed is wasted when self-fertilisation cannot occur. The selection gradient on sex allocation, meanwhile, is given by

$$S_1(\mathbf{z}) = K(\mathbf{z}) \left(\frac{1}{z_1} - \frac{1}{1 - z_1} \right), \quad (\text{A25c})$$

which cancels at $z_1^* = 1/2$, as expected in a randomly mating population.

Under enforced selfing ($z_3 = 1$), the selection gradient on pollen export is zero ($S_2(\mathbf{z}) = 0$) because pollen export does not affect reproductive success: pollen dispersal is neutral. The selection gradient on sex allocation is given by

$$S_1(\mathbf{z}) = K(\mathbf{z}) \frac{2(1 - \delta)}{z_1}, \quad (\text{A25d})$$

which is positive, indicating that selection favours allocating all resources to female function. This is because the amount of pollen produced does not matter for reproductive success as any non-zero production is sufficient for self-fertilisation, so that individuals producing more ovules are always at an advantage. Eventually, however, this should lead the population to produce no pollen at all and so to go extinct. This degenerate case stems from our assumption that pollen is never limiting for ovule fertilisation. In a more elaborate model of mating where the number of pollen grains required for fertilisation of available ovules is explicitly accounted for, the population would likely stabilise at a small but non-zero production of pollen, as seen in highly selfing species exhibiting a selfing syndrome.

A.3.2 Sexual interference caused by self-pollen promotion

The above shows that mixed mating is never favoured by directional selection when maternal plants can manipulate the outcome of pollen competition without incurring any cost, consistent with existing theory (e.g., Lande and Schemske, 1985; Abu Awad and Roze, 2020). We now investigate the effect of sexual interference on the joint evolution of self-pollen advantage, pollen export and sex allocation.

A.3.2.1 Directional selection

Under sexual interference, the selection gradient on self-pollen advantage is given by

$$S_3(\mathbf{z}) = K(\mathbf{z}) \underbrace{\frac{z_2(1-c)(1-z_2)P(z_3)}{[z_2(1-c)(1-z_3)P(z_3) + (1-z_2)z_3]^2}}_{\geq 0} \times \left[1 - 2\delta + (1-z_3) \left(z_3 + \frac{z_2(1-c)}{1-z_2}(1-z_3)P(z_3) \right) \frac{P'(z_3)}{P(z_3)} \right], \quad (\text{A26})$$

which does not depend on sex allocation, as before, and now features an extra term that depends on the sexual interference function $P(z_3)$ and its first derivative ($P'(z_3)$). This term is negative when self-pollen promotion causes sexual interference (i.e., when $P'(z_3) < 0$) and so opposes selection for self-fertilisation owing to the transmission advantage when inbreeding depression is weak ($\delta < 1/2$), indicating that sexual interference could in principle allow the existence of an intermediate equilibrium self-pollen advantage,

potentially leading to mixed mating.

The selection gradient on pollen export, meanwhile, is given by

$$S_2(\mathbf{z}) = K(\mathbf{z}) \underbrace{\frac{(1-c)(1-z_3)P(z_3)}{[z_2(1-c)(1-z_3)P(z_3) + (1-z_2)z_3]^2}}_{\geq 0} \left[z_3 - z_2 \left(2(1-\delta)z_3 - P(z_3)(1-c)(1-z_3) \right) \right]. \quad (\text{A27a})$$

The term on the right-hand side is always positive when $z_3 < 1$ and so plays no role in determining the direction of selection. The term on the second line is linear in z_2 . When $z_2 = 0$, the gradient reduces to

$$S_2(\mathbf{z}) = K(\mathbf{z}) \frac{P(z_3)(1-c)(1-z_3)}{z_3} \quad (\text{A27b})$$

which is always positive for $0 \leq z_3 < 1$, indicating that selection always favours some pollen export provided that outcrossing is possible. Therefore, if $S_2(\mathbf{z}) < 0$ when $z_2 = 1$, then there is a single singular pollen export value to which the population will converge under directional selection. Otherwise, selection favours complete pollen export ($z_2 \rightarrow 1$). When $z_2 = 1$, the gradient becomes

$$S_2(\mathbf{z}) = K(\mathbf{z}) \left[1 - \frac{z_3(1-2\delta)}{(1-c)(1-z_3)P(z_3)} \right], \quad (\text{A27c})$$

which is always positive when $\delta \geq 1/2$. Selection thus favours the evolution of complete pollen export ($z_2 \rightarrow 1$) when $\delta \geq 1/2$, that is, when inbreeding depression outweighs the transmission advantage of selfing. Otherwise, whether the gradient turns negative depends on the value of z_3 .

One can show that because $P'(z_3) < 0$, eq. (A27c) is a strictly decreasing function of z_3 for $\delta < 1/2$, is strictly positive when $z_3 = 0$ and tends to $-\infty$ as z_3 approaches 1, i.e.,

$$\lim_{z_3 \rightarrow 1^-} S_2(\mathbf{z}) = -\infty. \quad (\text{A27d})$$

These properties imply that the gradient on pollen export at $z_2 = 1$ is negative above a critical value $z'_3 \in [0, 1)$. Thus, when $z'_3 < z_3 < 1$ there is a singular intermediate pollen export rate $z_2^* \in (0, 1)$ and otherwise selection favours complete export ($z_2 \rightarrow 1$). When $z_3 > z'_3$, the singular pollen export rate is

given by

$$z_2^* = \frac{z_3}{2(1-\delta)z_3 - P(z_3)(1-c)(1-z_3)}. \quad (\text{A27e})$$

Overall, The analysis presented in (A27) shows that either complete pollen export evolves, or selection favours an intermediate pollen export rate (eq. A27e), which occurs when $\delta < 1/2$ and $z_3' < z_3 < 1$ (Note that when $z_3 = 1$, the pollen export rate is effectively neutral as only self-pollen can fertilise ovules).

Complete outcrossing is always stable. When $z_2 \rightarrow 1$, the selection gradient on self-pollen advantage (eq. A26) becomes

$$S_3(\mathbf{z}) = K(\mathbf{z}) \frac{P'(z_3)}{P(z_3)}, \quad (\text{A28})$$

which is strictly negative (since $P'(z_3) < 0$). This shows that in the presence of sexual interference, a population that is initially exporting pollen at a high rate will be driven towards a self-pollen advantage of zero. In the vicinity of $z_2 = 1$ and $z_3 = 0$, we thus have that $S_2(\mathbf{z}) = 1 > 0$ and $S_3(\mathbf{z}) < 0$, which shows that selection always maintains the population at this fully outcrossing strategy, making it an absorbing state of gradual evolutionary dynamics at which evolution comes to a halt, because directional selection does not vanish on either trait. The reason this holds even when inbreeding depression is absent is that the two traits interfere with one another. When z_3 is close to zero, reducing z_2 to retain pollen for self-fertilisation leads to such a small increase in selfing rate that this benefit is always trumped by the cost incurred in terms of lost mating success as pollen donor. When z_2 is close to one, meanwhile, increasing z_3 leads to a loss of pollen export under sexual interference that always outweighs the potential benefits in terms of increased selfing. Thus, enforced outcrossing is an evolutionary trap that cannot be escaped under sexual interference.

Complete selfing is also stable. When $z_3 = 1$, so that only self-pollen can fertilise maternal plants, the gradient on self-pollen advantage reduces to

$$S_3(\mathbf{z}) = K(\mathbf{z}) \frac{z_2(1-c)}{1-z_2} (1-2\delta)P(1), \quad (\text{A29})$$

which is strictly positive when $\delta < 1/2$, showing that z_3 is always selected to increase when it is sufficiently close to one under weak inbreeding depression. In the vicinity of $z_3 = 1$, the pollen export rate is also driven towards an intermediate z_2^* (eq. A27e). Thus, a population with an initially high self-pollen advantage will be selected to remain so, showing that complete selfing is also a stable equilibrium of directional selection under sexual interference.

Mixed mating can be favoured. One or more stable mixed mating equilibria could exist between the complete selfing and complete outcrossing equilibria, depending on the shape of the sexual interference function $P(z_3)$. However, explicitly showing the existence of such intermediate stable points becomes difficult even in the simplest of cases because eq.(A26) depends on z_3 in a complicated way. We therefore turn to numerical analyses, which requires picking a shape for $P(z_3)$. For the rest of our analyses, we assume that $P(z_3)$ is given by

$$P(z_3) = \frac{2(1 - e^{\kappa/2})}{2 - e^{\kappa/2} - e^{\kappa z_3}}. \quad (\text{A30})$$

Equation (A30) is a decreasing sigmoid function that satisfies $P(1/2) = 1$, so that our previous model without maternal control on pollen competition is recovered for $z_3 = 1/2$. The shape of $P(z_3)$ is controlled by parameter $\kappa > 0$. The higher κ , the more intense sexual interference is caused by self-pollen promotion.

We analysed the outcome of directional selection numerically using the approach described in section A.1.2.2, results are presented in Figure 2B-D in the main text.

A.3.2.2 Disruptive selection

The complete selfing and complete outcrossing equilibria are endpoints of gradual evolution that need not be given further consideration. In contrast, the mixed mating equilibrium (where it exists) is a singular point at which second-order forces may act. We computed the leading eigenvalue of the Hessian matrix $\mathbf{H}(\mathbf{z}^*)$ at this equilibrium and found that it is never evolutionarily stable, as the leading eigenvalue of $\mathbf{H}(\mathbf{z}^*)$ is always positive. See Main Text Section 4.3 for details.

Appendix B

Individual-based simulations

In this section, we describe the individual-based simulation program that we used to obtain the results presented in Figs. 1B and 3B-C of the main text. The program is coded in C++20 and is available at: [10.5281/zenodo.23062245](https://doi.org/10.5281/zenodo.23062245).

It depicts a population of N haploid individuals characterised by their allelic values for three traits: sex allocation z_1 , pollen export rate z_2 and self-pollen advantage z_3 . Each generation proceeds as follows.

Fecundities and selfing rates. We first determine the trait values expressed by each individual by clipping their allelic values so that they lie between 0 and 1. From these, we compute the female fecundity of individuals as

$$F(z_{1,i}) = z_{1,i} \quad (\text{B1})$$

for individual $i \in \{1, \dots, N\}$; their contribution to the outcross pollen pool, given by

$$M_{\text{out}}(\mathbf{z}_i) = (1 - z_{1,i})z_{2,i}P(z_{3,i}), \quad (\text{B2})$$

where

$$P(z_{3,i}) = \frac{2(1 - e^{\kappa/2})}{2 - e^{\kappa/2} - e^{\kappa z_{3,i}}} \quad (\text{B3})$$

denotes pollen discounting caused by the self-pollen advantage, and their selfing rate,

$$\alpha(\mathbf{z}_i) = \frac{z_{3,i}M_{\text{self}}(\mathbf{z}_i)}{z_{3,i}M_{\text{self}}(\mathbf{z}_i) + (1 - z_{3,i})\frac{1 - c}{N - 1} \sum_{j \neq i} M_{\text{out}}(\mathbf{z}_j)}, \quad (\text{B4})$$

where $M_{\text{self}}(\mathbf{z}_i) = (1 - z_{1,i})(1 - z_{2,i})$.

Offspring production. To create the next generation, we create N offspring as follows. For each offspring, we first pick a mother with replacement from the current generation. Each individual i has a probability $F(z_{1,i}) / \sum_j F(z_{1,j})$. We determine whether the produced offspring is selfed (with probability $\alpha(\mathbf{z}_i)$) or outcrossed (with probability $1 - \alpha(\mathbf{z}_i)$). If it is selfed, we mark it as such and assign it a viability $w = w_{\text{self}} = 1 - \delta$. If it is outcrossed, we also mark it and assign it a viability $w = w_{\text{out}} = 1$. We then check whether the produced offspring survives (with probability w). If it does not survive, we discard it and start over by sampling a new mother, determining whether the new offspring is selfed and so on. We repeat this process until a surviving offspring is obtained.

Haplotype transmission. If this surviving offspring is selfed, then it inherits its mother's haplotype. If it is outcrossed, it either inherits the mother's haplotype (with probability $1/2$) or that of its father, in which case said father must be sampled from the population. To do so, we pick individuals from the current generation, with the probability that individual i is chosen given by $M_{\text{out}}(\mathbf{z}_i) / \sum_j M_{\text{out}}(\mathbf{z}_j)$. If the chosen individual turns out to be the same as the mother, it is discarded and a new father is picked until we find one that is different from the maternal individual.

Mutation. Once the offspring has been constructed, it undergoes mutation with probability μ , in which case all the allelic values are changed by a small deviation sampled independently in a normal distribution with mean zero and standard deviation σ . Allelic values are clamped to remain between -0.2 and 1.2 . Allowing allelic values to vacate the $[0, 1]$ interval over which traits are defined prevents the occurrence of directional mutational bias due to boundary effects.

Measurements. Every t_{mes} generations, we record the allelic values for all traits and the selfing rates of n_{mes} randomly sampled individuals in the population.

Initialisation. The population is initially fixed for phenotype $\mathbf{z}_{\text{init}} = (z_{1,\text{init}}, z_{2,\text{init}}, z_{3,\text{init}})$.

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